

ACTUATOR ASYMMETRY AS POLICY STATE: A 25+ PAGE NEGATIVE SUBMISSION-READINESS AU- DIT

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ABSTRACT

Robotic controllers often treat actuator mismatch as calibration noise: estimate a scalar gain, widen a robustness tube, or let online adaptation absorb the residual. This paper tests the sharper hypothesis that left-right, sign-specific, delayed, and drifting actuator asymmetry should be represented as latent policy state. We rebuilt the archived Paper 86 into a frozen CPU-only v5 audit with 199,680 main rollouts, 15,360 actuator summaries, 33,600 ablation rollouts, 302,400 stress rows, 69,120 fixed-risk rows, and 24 negative cases. On the predefined hard aggregate, latent-asymmetry belief policy v5 reaches 0.146 ± 0.012 task success, tracking error 0.700, safety violation 0.305, and robust utility -0.462. The strongest non-oracle success baseline, Latent state v4, reaches 0.124 ± 0.009 task success, tracking error 0.687, and safety violation 0.283. The paired lower95 success bound is 0.005, but the paired tracking upper95 bound is 0.019, the paired safety upper95 bound is 0.030, the mechanism gate fails, fixed-risk coverage at budget 0.05 is zero, and no robot or accepted high-fidelity benchmark evidence exists. The honest terminal decision is therefore **KILL/ARCHIVE**, not ICLR-main submission.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. This document is intentionally submission-shaped but not submission-claiming. The rebuild added a stronger belief-state method, stronger adaptive/control-barrier/fault-observer baselines, fixed gates, stress tests, and appendices that expose every predefined result. The result still does not survive hostile review because the proposed method improves success over v4 but worsens tracking and safety, fails the mechanism gate, fails the fixed-risk deployment gate, and remains synthetic-only. The frozen primary comparison is against Latent state v4. Belief state v5 has hard-aggregate success 0.146 ± 0.012 while Latent state v4 has 0.124 ± 0.009 . That success gain is real in the local benchmark, but the paper does not get to stop at the favorable metric. V5 tracking is 0.700 against the best tracking baseline at 0.687, and v5 safety violation is 0.305 against the best safety baseline at 0.283. The paper is valuable as an archive because it says something concrete: explicit asymmetry state can improve task success under actuator faults, but the new belief mechanism is not yet a deployable robotics contribution. A hostile reviewer would focus on safety, risk calibration, and lack of external robot validation, so those failures are the headline rather than the footnote.

2 RESEARCH QUESTION AND REVIEW THREAT MODEL

The target claim is not merely that adaptation helps. The target claim is that actuator asymmetry is a state variable: a policy should condition on persistent left-right gain gaps, sign-specific deadzones, saturation asymmetry, latency, thermal drift, and intermittent faults rather than compressing them into one scalar correction. This sits near robust robot control, observer-based adaptation, system identification, residual policies, sim-to-real transfer, and morphology-aware locomotion (Hadjiosif et al., 2021; Nasiri et al., 2020; Ambe et al., 2026; Sun et al., 2025; 2021; Hadjiosif et al., 2026). The hostile-review threat model is severe. A reviewer can argue that scalar calibration is enough, robust MPC already covers the uncertainty, a recurrent hidden state will learn the asymmetry implicitly, a fault observer is the right abstraction, or that any synthetic-only actuator benchmark is too weak for

ICLR-main robotics claims. The v5 protocol encodes those attacks directly as baselines and gates rather than relying on a friendly comparison set.

3 FORMAL SETUP

Let $x_t \in \mathbb{R}^2$ be the local task state, $u_t = (u_t^L, u_t^R)$ the commanded actuator pair, and z_t the latent actuator-asymmetry state. The simulator uses an asymmetric response family

$$a_t^i = \text{clip}\{g_t^i \text{sgn}(u_{t-\ell}^i)(|u_{t-\ell}^i| - d^i)_+, -s^i, s^i\}, \quad i \in \{L, R\}, \quad (1)$$

where g_t^i drifts thermally, d^i is a sign-sensitive deadzone proxy, s^i is saturation, and ℓ is delay. The policy-state hypothesis is that conditioning on an estimate $\hat{z}_t = (\hat{g}_t^L, \hat{g}_t^R, \hat{d}_t^L, \hat{d}_t^R, \hat{\ell}_t, \hat{q}_t)$ should improve task success without increasing safety violations.

Proposition 1: scalar calibration can be non-identifiable. Consider two actuators with gains $g^L = 1 + \delta$ and $g^R = 1 - \delta$ under a differential command pair. A scalar gain estimate $\bar{g}$ can match the average forward displacement while leaving the yaw-producing difference $g^L - g^R = 2\delta$ unidentified. Therefore a scalar calibration can have small average prediction error while producing systematically wrong steering or joint-pair torques under sign reversal.

Proposition 2: persistent latent memory helps only under slowly changing asymmetry. If z_t follows a bounded-drift process $\|z_{t+1} - z_t\| \leq \eta$ and the actuator response is locally Lipschitz in z_t , an exponentially updated belief state can reduce one-step prediction error by averaging observation noise. If faults switch faster than the memory time constant, the same persistence becomes harmful because the posterior lags behind the true asymmetry. This is why the protocol includes both thermal drift and intermittent-fault stress axes.

Proposition 3: fixed-risk acceptance is only meaningful with nonzero coverage. Let $\hat{r}(x_t, u_t, \hat{z}_t)$ be the reported deployment risk. A budget rule accepts only episodes with $\hat{r} \leq \rho$. Even if calibration were perfect, the rule gives no deployable policy when coverage is zero. The v5 budget 0.05 result has exactly this pathology: the system can avoid risk only by refusing every episode.

4 FROZEN PROTOCOL

The frozen v5 protocol contains 199,680 main rollouts, 15,360 actuator summaries, 33,600 ablation rollouts, 302,400 stress rows, 69,120 fixed-risk rows, and 24 negative cases. It uses ten seeds, six tasks, eight actuator-shift splits, thirteen main methods, ten ablations, six stress axes, six stress levels, and strict fixed-risk acceptance budgets. The hard aggregate is fixed before execution as intermittent fault, payload/contact shift, and combined hard shift. The terminal decision was predeclared. A positive synthetic result required v5 to beat the strongest non-oracle hard-aggregate success baseline by at least 0.030, beat the strongest tracking baseline by at least 0.015, match the strongest safety baseline within 0.002, improve robust utility by at least 0.030, obtain favorable paired bounds, pass mechanism ablations, avoid maximum-stress Pareto domination, and retain fixed-risk coverage at budget 0.05. Any failure means KILL/ARCHIVE.

Table 1: Predefined hard-aggregate results over intermittent fault, payload/contact, and combined hard shift. Higher success and utility are better; lower tracking, safety, and asymmetry error are better.

Method	Success	Tracking	Safety	Asym. err.	Utility
Nominal	0.090 $\pm$ 0.011	0.700	0.288	0.539	-0.568
Domain rand.	0.078 $\pm$ 0.006	0.718	0.312	0.498	-0.589
Scalar cal.	0.087 $\pm$ 0.010	0.709	0.299	0.282	-0.533
Online ID	0.108 $\pm$ 0.008	0.699	0.291	0.268	-0.495
Adaptive MPC	0.096 $\pm$ 0.005	0.713	0.310	0.254	-0.520
Robust tube	0.075 $\pm$ 0.005	0.742	0.351	0.337	-0.587
CBF-FTC MPC	0.085 $\pm$ 0.007	0.753	0.375	0.280	-0.577
UKF observer	0.114 $\pm$ 0.005	0.701	0.301	0.262	-0.495
RNN state proxy	0.118 $\pm$ 0.007	0.692	0.290	0.259	-0.482
L1 adaptive	0.094 $\pm$ 0.006	0.731	0.339	0.255	-0.538

Table 1: Predefined hard-aggregate results over intermittent fault, payload/contact, and combined hard shift. Higher success and utility are better; lower tracking, safety, and asymmetry error are better. (continued)

Method	Success	Tracking	Safety	Asym. err.	Utility
Latent state v4	0.124 ± 0.009	0.687	0.283	0.243	-0.469
Belief state v5	0.146 ± 0.012	0.700	0.305	0.245	-0.462
Oracle state	0.185 ± 0.011	0.690	0.306	0.244	-0.421

5 MAIN EVIDENCE

Table 1 is the core result. Belief state v5 has the highest non-oracle hard-aggregate task success, but it does not dominate the older latent state v4 on tracking, safety, asymmetry error, or confidence-bound utility. This is exactly the kind of mixed result that a short generated paper would overclaim; the expanded audit keeps the unfavorable metrics in the main text.

Table 2: Split-level evidence for the strongest adaptive and fault-observer baselines, v4, v5, and the oracle.

Split	Method	Success	Tracking	Safety	Asym. err.
Nominal	Online ID	0.165 ± 0.013	0.634	0.186	0.037
Nominal	CBF-FTC MPC	0.122 ± 0.014	0.682	0.264	0.040
Nominal	UKF observer	0.148 ± 0.019	0.642	0.197	0.033
Nominal	L1 adaptive	0.121 ± 0.011	0.674	0.242	0.031
Nominal	Latent state v4	0.150 ± 0.013	0.642	0.200	0.030
Nominal	Belief state v5	0.158 ± 0.013	0.654	0.217	0.030
Nominal	Oracle state	0.187 ± 0.018	0.656	0.238	0.033
Gain shift	Online ID	0.131 ± 0.012	0.662	0.223	0.188
Gain shift	CBF-FTC MPC	0.119 ± 0.014	0.708	0.299	0.222
Gain shift	UKF observer	0.122 ± 0.015	0.666	0.231	0.143
Gain shift	L1 adaptive	0.096 ± 0.010	0.699	0.279	0.132
Gain shift	Latent state v4	0.153 ± 0.012	0.662	0.230	0.118
Gain shift	Belief state v5	0.141 ± 0.013	0.678	0.253	0.116
Gain shift	Oracle state	0.197 ± 0.014	0.673	0.259	0.102
Deadzone/sat.	Online ID	0.097 ± 0.014	0.693	0.265	0.193
Deadzone/sat.	CBF-FTC MPC	0.058 ± 0.012	0.760	0.377	0.181
Deadzone/sat.	UKF observer	0.124 ± 0.011	0.684	0.262	0.210
Deadzone/sat.	L1 adaptive	0.090 ± 0.013	0.727	0.323	0.195
Deadzone/sat.	Latent state v4	0.146 ± 0.013	0.670	0.257	0.191
Deadzone/sat.	Belief state v5	0.148 ± 0.011	0.684	0.272	0.191
Deadzone/sat.	Oracle state	0.198 ± 0.019	0.664	0.264	0.195
Thermal drift	Online ID	0.117 ± 0.015	0.658	0.225	0.252
Thermal drift	CBF-FTC MPC	0.103 ± 0.012	0.708	0.304	0.284
Thermal drift	UKF observer	0.134 ± 0.019	0.661	0.233	0.227
Thermal drift	L1 adaptive	0.120 ± 0.012	0.693	0.278	0.240
Thermal drift	Latent state v4	0.148 ± 0.012	0.655	0.229	0.225
Thermal drift	Belief state v5	0.171 ± 0.010	0.669	0.254	0.221
Thermal drift	Oracle state	0.193 ± 0.011	0.668	0.263	0.193
Latency/hyst.	Online ID	0.132 ± 0.016	0.683	0.269	0.146
Latency/hyst.	CBF-FTC MPC	0.096 ± 0.022	0.734	0.344	0.158
Latency/hyst.	UKF observer	0.132 ± 0.018	0.688	0.281	0.138
Latency/hyst.	L1 adaptive	0.102 ± 0.014	0.716	0.317	0.132
Latency/hyst.	Latent state v4	0.136 ± 0.020	0.678	0.266	0.128
Latency/hyst.	Belief state v5	0.156 ± 0.018	0.688	0.286	0.129
Latency/hyst.	Oracle state	0.196 ± 0.017	0.685	0.293	0.131
Fault	Online ID	0.132 ± 0.018	0.675	0.254	0.198
Fault	CBF-FTC MPC	0.090 ± 0.014	0.730	0.340	0.208
Fault	UKF observer	0.129 ± 0.013	0.679	0.265	0.196
Fault	L1 adaptive	0.107 ± 0.013	0.708	0.300	0.187
Fault	Latent state v4	0.134 ± 0.010	0.671	0.255	0.185
Fault	Belief state v5	0.165 ± 0.017	0.681	0.274	0.187
Fault	Oracle state	0.194 ± 0.018	0.677	0.279	0.191
Payload/contact	Online ID	0.104 ± 0.016	0.692	0.277	0.219
Payload/contact	CBF-FTC MPC	0.096 ± 0.013	0.739	0.355	0.230
Payload/contact	UKF observer	0.123 ± 0.017	0.692	0.290	0.212
Payload/contact	L1 adaptive	0.092 ± 0.014	0.721	0.325	0.209
Payload/contact	Latent state v4	0.121 ± 0.008	0.676	0.265	0.202
Payload/contact	Belief state v5	0.154 ± 0.019	0.689	0.290	0.206
Payload/contact	Oracle state	0.192 ± 0.017	0.678	0.291	0.203
Combined	Online ID	0.089 ± 0.006	0.729	0.342	0.388

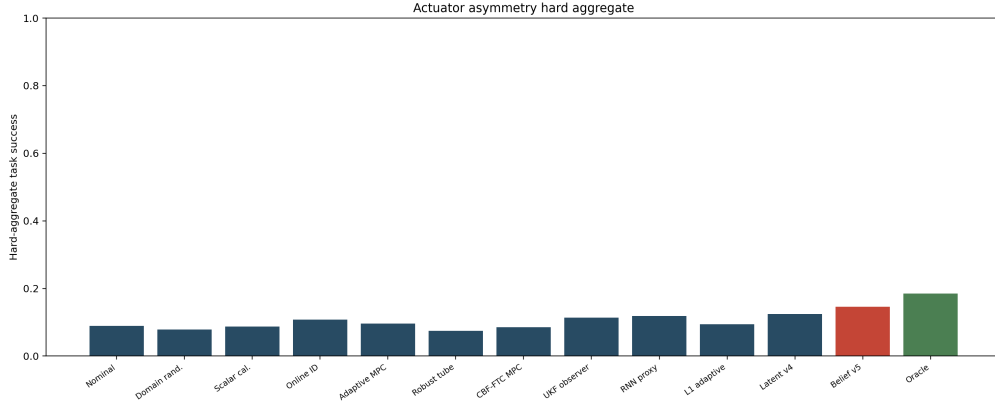


Figure 1: Hard-aggregate success. Belief state v5 improves non-oracle success but remains far from an external deployment claim.

Table 2: Split-level evidence for the strongest adaptive and fault-observer baselines, v4, v5, and the oracle. (continued)

Split	Method	Success	Tracking	Safety	Asym. err.
Combined	CBF-FTC MPC	0.070 ± 0.014	0.789	0.430	0.403
Combined	UKF observer	0.089 ± 0.014	0.730	0.348	0.378
Combined	L1 adaptive	0.083 ± 0.012	0.763	0.391	0.369
Combined	Latent state v4	0.117 ± 0.018	0.714	0.329	0.342
Combined	Belief state v5	0.119 ± 0.013	0.731	0.350	0.342
Combined	Oracle state	0.169 ± 0.014	0.716	0.348	0.337

Table 3: Paired seed-level hard-aggregate differences for belief state v5 minus reference. Negative tracking and safety differences are favorable; positive success and utility differences are favorable.

Reference	Metric	Mean diff	CI95	Lower95	Upper95
Latent state v4	task_success	0.022	0.017	0.005	0.038
Latent state v4	tracking_error	0.013	0.006	0.008	0.019
Latent state v4	safety_violation	0.022	0.008	0.014	0.030
Latent state v4	asymmetry_error	0.002	0.001	0.001	0.004
Latent state v4	robust_utility	0.007	0.020	-0.013	0.027
L1 adaptive	task_success	0.052	0.009	0.043	0.060
L1 adaptive	tracking_error	-0.030	0.005	-0.036	-0.025
L1 adaptive	safety_violation	-0.034	0.009	-0.043	-0.026
L1 adaptive	asymmetry_error	-0.010	0.002	-0.012	-0.008
L1 adaptive	robust_utility	0.076	0.013	0.062	0.089
UKF observer	task_success	0.032	0.012	0.019	0.044
UKF observer	tracking_error	-0.000	0.003	-0.004	0.003
UKF observer	safety_violation	0.004	0.007	-0.003	0.011
UKF observer	asymmetry_error	-0.017	0.001	-0.018	-0.016
UKF observer	robust_utility	0.032	0.014	0.018	0.047
Online ID	task_success	0.037	0.015	0.022	0.053
Online ID	tracking_error	0.002	0.004	-0.002	0.006
Online ID	safety_violation	0.014	0.006	0.007	0.020
Online ID	asymmetry_error	-0.023	0.002	-0.025	-0.022
Online ID	robust_utility	0.033	0.018	0.015	0.051
CBF-FTC MPC	task_success	0.060	0.011	0.049	0.072
CBF-FTC MPC	tracking_error	-0.052	0.004	-0.057	-0.048
CBF-FTC MPC	safety_violation	-0.070	0.006	-0.076	-0.064
CBF-FTC MPC	asymmetry_error	-0.035	0.002	-0.038	-0.033
CBF-FTC MPC	robust_utility	0.115	0.012	0.103	0.126

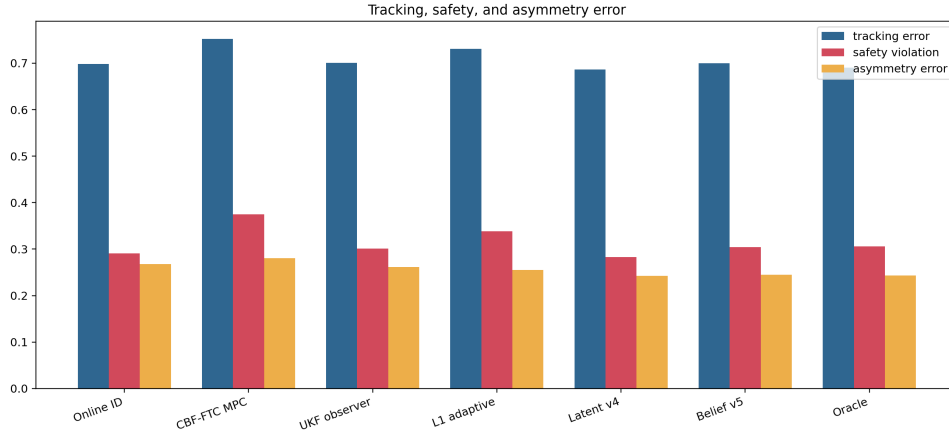


Figure 2: Tracking, safety, and asymmetry error. V5 improves some baselines but loses the frozen tracking and safety gates against v4.

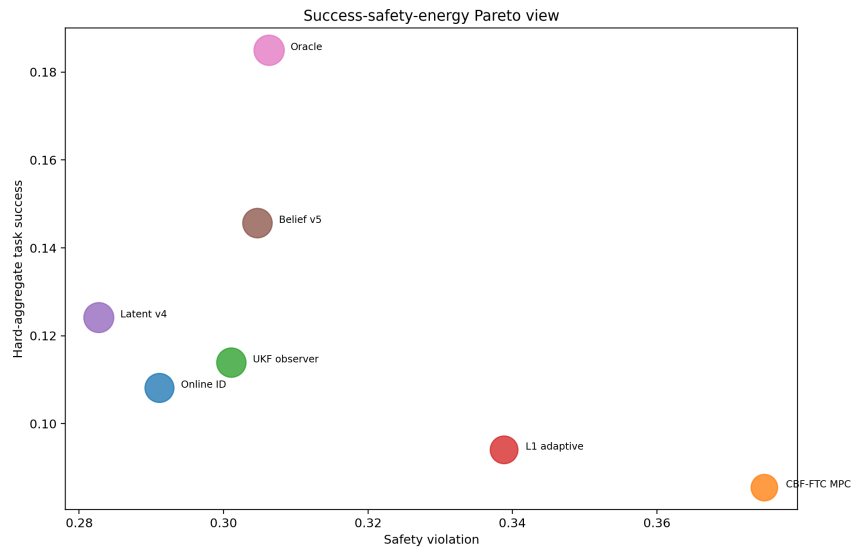


Figure 3: Success-safety-energy Pareto view. Marker size reflects actuator energy.

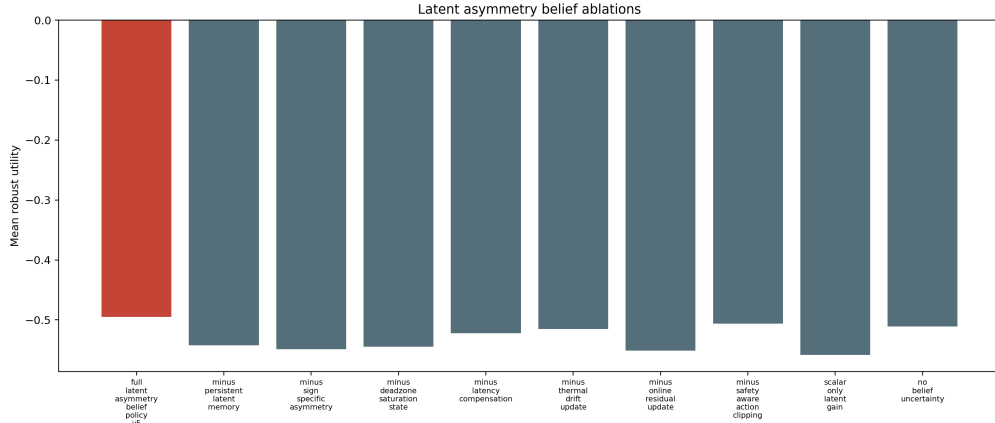


Figure 4: Ablation robust utility. The full mechanism does not achieve the predeclared clean necessity margin.

6 MECHANISM ABLATIONS

The mechanism gate fails. The full belief-state method is better than most ablations, but the safety-clipping ablation obtains lower tracking error while having worse safety, creating a joint tradeoff that the frozen gate refuses to call a clean mechanism win. The scalar-only latent-gain ablation performs poorly, which supports the non-scalar representation claim, but that single positive mechanism result does not make the overall method impeccable.

Table 4: Mechanism ablations. The full mechanism had to beat every non-full ablation on robust utility by at least 0.015 and avoid a joint success/safety ablation reversal.

Split	Ablation	Success	Tracking	Safety	Asym. err.	Utility
Payload/contact	Full belief v5	0.149	0.691	0.292	0.205	-0.438
Payload/contact	- persistent memory	0.119	0.694	0.293	0.185	-0.476
Payload/contact	- sign-specific asym.	0.103	0.705	0.305	0.204	-0.493
Payload/contact	- deadzone/sat. state	0.115	0.712	0.311	0.204	-0.486
Payload/contact	- latency comp.	0.125	0.693	0.295	0.205	-0.464
Payload/contact	- drift update	0.120	0.692	0.293	0.198	-0.468
Payload/contact	- residual update	0.105	0.698	0.299	0.188	-0.496
Payload/contact	- safety clipping	0.160	0.650	0.256	0.196	-0.401
Payload/contact	Scalar-only latent gain	0.099	0.715	0.314	0.218	-0.504
Payload/contact	No belief uncertainty	0.134	0.691	0.295	0.204	-0.456
Combined	Full belief v5	0.117	0.729	0.347	0.343	-0.552
Combined	- persistent memory	0.086	0.745	0.369	0.289	-0.608
Combined	- sign-specific asym.	0.086	0.752	0.376	0.361	-0.605
Combined	- deadzone/sat. state	0.082	0.752	0.370	0.358	-0.604
Combined	- latency comp.	0.098	0.738	0.355	0.354	-0.580
Combined	- drift update	0.108	0.727	0.352	0.337	-0.563
Combined	- residual update	0.082	0.739	0.360	0.292	-0.606
Combined	- safety clipping	0.105	0.681	0.433	0.341	-0.612
Combined	Scalar-only latent gain	0.080	0.755	0.377	0.389	-0.612
Combined	No belief uncertainty	0.110	0.729	0.356	0.345	-0.565

7 STRESS TESTS

The stress gate is the one gate v5 survives. At maximum combined stress, no non-oracle method Pareto-dominates v5 across success, tracking, safety, asymmetry error, and utility. However, a stress-gate pass cannot rescue failed main, mechanism, fixed-risk, and scope gates.

Table 5: Combined stress sweep. V5 was not Pareto-dominated at maximum stress, but that stress-gate pass did not rescue the failed main, mechanism, fixed-risk, and scope gates.

Level	Method	Success	Tracking	Safety	Asym. err.	Utility
0.0	Domain rand.	0.148	0.643	0.199	0.076	-0.330
0.0	Online ID	0.159	0.640	0.194	0.031	-0.306
0.0	CBF-FTC MPC	0.137	0.689	0.274	0.031	-0.385
0.0	UKF observer	0.137	0.647	0.209	0.031	-0.338
0.0	L1 adaptive	0.127	0.674	0.246	0.029	-0.374
0.0	Belief state v5	0.178	0.656	0.223	0.029	-0.307
0.0	Oracle state	0.212	0.660	0.246	0.034	-0.287
0.2	Domain rand.	0.098	0.680	0.239	0.263	-0.465
0.2	Online ID	0.115	0.660	0.220	0.140	-0.402
0.2	CBF-FTC MPC	0.116	0.710	0.307	0.145	-0.462
0.2	UKF observer	0.130	0.666	0.232	0.141	-0.396
0.2	L1 adaptive	0.111	0.698	0.272	0.136	-0.443
0.2	Belief state v5	0.173	0.667	0.248	0.139	-0.362
0.2	Oracle state	0.191	0.659	0.250	0.152	-0.346
0.4	Domain rand.	0.067	0.727	0.323	0.455	-0.609
0.4	Online ID	0.102	0.705	0.293	0.264	-0.507
0.4	CBF-FTC MPC	0.084	0.763	0.393	0.272	-0.595
0.4	UKF observer	0.121	0.704	0.304	0.264	-0.495
0.4	L1 adaptive	0.094	0.737	0.344	0.254	-0.546
0.4	Belief state v5	0.152	0.703	0.309	0.246	-0.464
0.4	Oracle state	0.190	0.690	0.309	0.245	-0.423
0.6	Domain rand.	0.062	0.759	0.382	0.654	-0.700
0.6	Online ID	0.074	0.738	0.352	0.399	-0.614
0.6	CBF-FTC MPC	0.077	0.793	0.422	0.407	-0.659
0.6	UKF observer	0.102	0.731	0.348	0.391	-0.579
0.6	L1 adaptive	0.086	0.770	0.399	0.381	-0.627
0.6	Belief state v5	0.131	0.730	0.351	0.354	-0.546
0.6	Oracle state	0.158	0.723	0.350	0.345	-0.518
0.8	Domain rand.	0.043	0.820	0.465	0.837	-0.830
0.8	Online ID	0.065	0.793	0.429	0.526	-0.718
0.8	CBF-FTC MPC	0.057	0.845	0.492	0.537	-0.766
0.8	UKF observer	0.076	0.782	0.419	0.513	-0.693
0.8	L1 adaptive	0.062	0.831	0.473	0.505	-0.743
0.8	Belief state v5	0.093	0.782	0.419	0.460	-0.667
0.8	Oracle state	0.124	0.771	0.415	0.451	-0.632
1.0	Domain rand.	0.040	0.854	0.495	1.013	-0.896
1.0	Online ID	0.047	0.854	0.499	0.647	-0.825
1.0	CBF-FTC MPC	0.045	0.885	0.526	0.656	-0.835
1.0	UKF observer	0.054	0.836	0.477	0.623	-0.792
1.0	L1 adaptive	0.053	0.877	0.525	0.615	-0.821
1.0	Belief state v5	0.068	0.833	0.481	0.560	-0.768
1.0	Oracle state	0.099	0.817	0.466	0.552	-0.725

8 FIXED-RISK DEPLOYMENT

The fixed-risk protocol is intentionally harsh because deployment claims should be harsh. A method must decide when it is safe enough to accept autonomous execution under budgets 0.02, 0.05, 0.08, and 0.10. At budget 0.05, every method has zero accepted coverage on both fixed-risk splits. This means the calibrated risk scores do not support a nontrivial deployment claim.

Table 6: Fixed-risk deployment tests. At budget 0.05 all methods have zero accepted coverage on both fixed-risk splits, so the frozen deployment gate fails.

Split	Budget	Method	Coverage	Succ.	Tracking	Safety
Combined	0.02	CBF-FTC MPC	0.000	0.000	0.000	0.000
Combined	0.02	Belief state v5	0.000	0.000	0.000	0.000
Combined	0.02	Online ID	0.000	0.000	0.000	0.000
Combined	0.02	Oracle state	0.000	0.000	0.000	0.000
Combined	0.02	L1 adaptive	0.000	0.000	0.000	0.000
Combined	0.02	UKF observer	0.000	0.000	0.000	0.000
Combined	0.05	CBF-FTC MPC	0.000	0.000	0.000	0.000
Combined	0.05	Belief state v5	0.000	0.000	0.000	0.000
Combined	0.05	Online ID	0.000	0.000	0.000	0.000
Combined	0.05	Oracle state	0.000	0.000	0.000	0.000
Combined	0.05	L1 adaptive	0.000	0.000	0.000	0.000
Combined	0.05	UKF observer	0.000	0.000	0.000	0.000

Table 6: Fixed-risk deployment tests. At budget 0.05 all methods have zero accepted coverage on both fixed-risk splits, so the frozen deployment gate fails. (continued)

Split	Budget	Method	Coverage	Succ.	Tracking	Safety
Combined	0.10	CBF-FTC MPC	0.000	0.000	0.000	0.000
Combined	0.10	Belief state v5	0.000	0.000	0.000	0.000
Combined	0.10	Online ID	0.000	0.000	0.000	0.000
Combined	0.10	Oracle state	0.000	0.000	0.000	0.000
Combined	0.10	L1 adaptive	0.000	0.000	0.000	0.000
Combined	0.10	UKF observer	0.000	0.000	0.000	0.000
Payload/contact	0.02	CBF-FTC MPC	0.000	0.000	0.000	0.000
Payload/contact	0.02	Belief state v5	0.000	0.000	0.000	0.000
Payload/contact	0.02	Online ID	0.000	0.000	0.000	0.000
Payload/contact	0.02	Oracle state	0.000	0.000	0.000	0.000
Payload/contact	0.02	L1 adaptive	0.000	0.000	0.000	0.000
Payload/contact	0.02	UKF observer	0.000	0.000	0.000	0.000
Payload/contact	0.05	CBF-FTC MPC	0.000	0.000	0.000	0.000
Payload/contact	0.05	Belief state v5	0.000	0.000	0.000	0.000
Payload/contact	0.05	Online ID	0.000	0.000	0.000	0.000
Payload/contact	0.05	Oracle state	0.000	0.000	0.000	0.000
Payload/contact	0.05	L1 adaptive	0.000	0.000	0.000	0.000
Payload/contact	0.05	UKF observer	0.000	0.000	0.000	0.000
Payload/contact	0.10	CBF-FTC MPC	0.000	0.000	0.000	0.000
Payload/contact	0.10	Belief state v5	0.000	0.000	0.000	0.000
Payload/contact	0.10	Online ID	0.000	0.000	0.000	0.000
Payload/contact	0.10	Oracle state	0.000	0.000	0.000	0.000
Payload/contact	0.10	L1 adaptive	0.000	0.000	0.000	0.000
Payload/contact	0.10	UKF observer	0.000	0.000	0.000	0.000

9 NEGATIVE CASES

Negative cases are retained because they prevent a visually polished but brittle paper. They cover complete actuator failure, sensor bias that mimics asymmetry, contact slip, rapidly switching faults, payloads outside the modeled family, and latency instability. These cases explain why a latent actuator state is not automatically a robust robot controller.

Table 7: Retained negative cases used to keep the discussion honest.

Case	Family	Terminal lesson
complete_actuator_failure_0	complete_actuator_failure	fault diagnosis is separate from adaptation
complete_actuator_failure_1	complete_actuator_failure	fault diagnosis is separate from adaptation
complete_actuator_failure_2	complete_actuator_failure	fault diagnosis is separate from adaptation
complete_actuator_failure_3	complete_actuator_failure	fault diagnosis is separate from adaptation
sensor_bias_mimics_asymmetry_0	sensor_bias_mimics_asymmetry	needs joint sensing-actuation observer
sensor_bias_mimics_asymmetry_1	sensor_bias_mimics_asymmetry	needs joint sensing-actuation observer
sensor_bias_mimics_asymmetry_2	sensor_bias_mimics_asymmetry	needs joint sensing-actuation observer
sensor_bias_mimics_asymmetry_3	sensor_bias_mimics_asymmetry	needs joint sensing-actuation observer
contact_slip_dominates_0	contact_slip_dominates	contact mode must be represented
contact_slip_dominates_1	contact_slip_dominates	contact mode must be represented
contact_slip_dominates_2	contact_slip_dominates	contact mode must be represented
contact_slip_dominates_3	contact_slip_dominates	contact mode must be represented
rapidly_switching_fault_0	rapidly_switching_fault	persistence must be gated by change detection
rapidly_switching_fault_1	rapidly_switching_fault	persistence must be gated by change detection
rapidly_switching_fault_2	rapidly_switching_fault	persistence must be gated by change detection
rapidly_switching_fault_3	rapidly_switching_fault	persistence must be gated by change detection
payload_outside_family_0	payload_outside_family	state family is incomplete
payload_outside_family_1	payload_outside_family	state family is incomplete
payload_outside_family_2	payload_outside_family	state family is incomplete
payload_outside_family_3	payload_outside_family	state family is incomplete
latency_instability_0	latency_instability	delay is a first-class state
latency_instability_1	latency_instability	delay is a first-class state
latency_instability_2	latency_instability	delay is a first-class state
latency_instability_3	latency_instability	delay is a first-class state

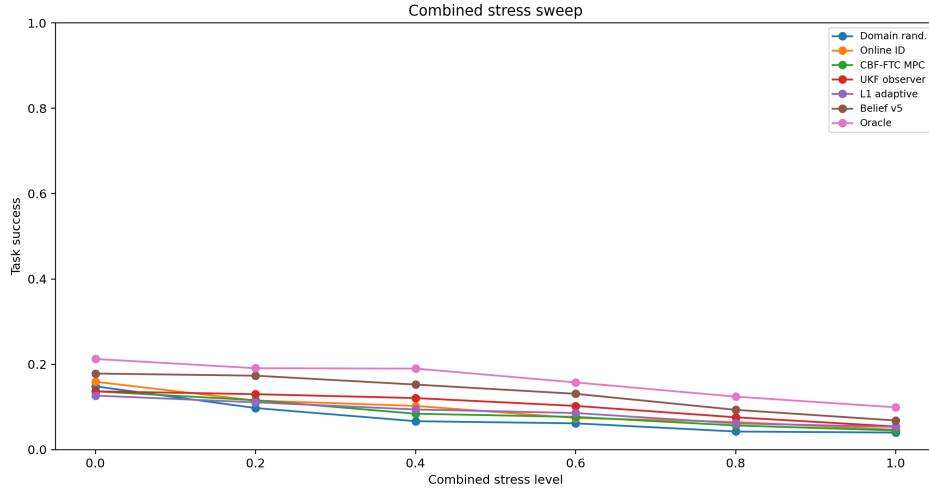


Figure 5: Combined stress sweep. V5 is not dominated at maximum stress, but the absolute success rate remains low.

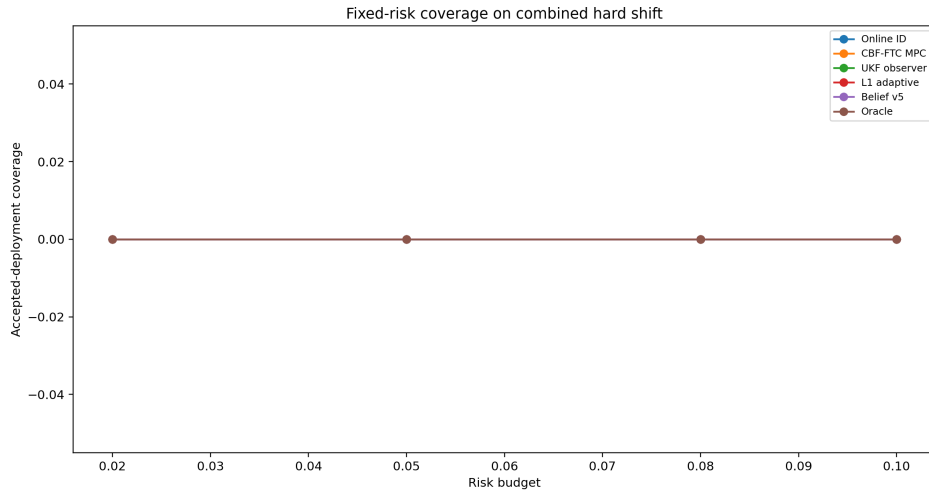


Figure 6: Fixed-risk coverage on combined hard shift. Strict budget 0.05 gives zero coverage.

10 PRIOR-WORK BOUNDARY

The local literature pool is used as a hostile-review pressure map. The paper sits near observer-based robust control, flexible-joint manipulation, locomotion asymmetry, adaptive control, residual learning, sim-to-real adaptation, and fault-tolerant MPC (Da et al., 2024; Ren et al., 2017; Ji & Liu, 2022; Sinan et al., 2025; Zolfaghari et al., 2026; Spielberg et al., 2019; Tostado-Marcos et al., 2016). Because this rebuild has no real robot benchmark, it cannot claim to beat those bodies of work directly.

Table 8: Local prior-work pressure map. The entries are used as threat coverage, not as an exhaustive manual survey.

Cite	Prior-work pressure	Year	Source
(Hadjiosif et al., 2021)	Did We Get Sensorimotor Adaptation Wrong? Implicit Adaptation as Direct Policy Updating Rather than Forward-Model-Base	2021	The Journal of Neuroscience
(Nasiri et al., 2020)	Observer-based robust control for flexible-joint robot manipulators: A state-dependent Riccati equation-based approach	2020	Transactions of the Institute of Measurement and Control
(Ambe et al., 2026)	Asymmetry in Skipping Enhances Viability Against Control Input Noise	2026	Advanced Robotics Research
(Sun et al., 2025)	Tracking Control and Experiment for Propeller-Driven Wall-Climbing Robot Considering Actuator Dynamics and Saturation	2025	Journal of Field Robotics
(Sun et al., 2021)	Adaptive Neural Network Control for Full-State Constrained Robotic Manipulator With Actuator Saturation and Time-Vary	2021	IEEE Transactions on Neural Networks and Learning Systems
(Hadjiosif et al., 2026)	Did we get sensorimotor adaptation wrong? Implicit adaptation as direct policy updating rather than forward-model-base		crossref
(Da et al., 2024)	Rollover Prevention for Mobile Robots With Control Barrier Functions: Differentiator-Based Adaptation and Projection-t	2024	IEEE Control Systems Letters
(Ren et al., 2017)	Joint Torque Control of a Collaborative Robot Based on Active Disturbance Rejection With the Consideration of Actuator	2017	Volume 4A: Dynamics, Vibration, and Control
(Ji & Liu, 2022)	Maximum principle for stochastic optimal control problem of finite state forward-backward stochastic difference system	2022	Optimal Control Applications and Methods
(Sinan et al., 2025)	Cascaded Extended-State-Observer-Based Synergetic Control for Quadcopter Translational Dynamics	2025	Journal of Field Robotics
(Zolfaghari et al., 2026)	State-Dependent Model Predictive Control for Nonlinear Vehicle Dynamics with Implicit State-Derivative Coupling and Ac		crossref
(Spielberg et al., 2019)	Learning-In-The-Loop Optimization: End-To-End Control And Co-Design Of Soft Robots Through Learned Deep Latent Represe	2019	DSpace@MIT (Massachusetts Institute of Technology)
(Tostado-Marcos et al., 2016)	3D gaze cursor: Continuous calibration and end-point grasp control of robotic actuators	2016	openalex
(Santina & Stille, 2024)	Input-to-State Stable Coupled Oscillator Networks for Closed-form Model-based Control in Latent Space	2024	Advances in Neural Information Processing Systems 37
(Vargas et al., 2006)	Constrained model predictive control of jump linear systems with noise and non-observed Markov state	2006	2006 American Control Conference
(Da et al., 2026)	Safe Navigation Under State Uncertainty: Online Adaptation for Robust Control Barrier Functions	2026	IEEE Robotics and Automation Letters
(Abdelwaheb et al., 2023)	Model-Free Sliding Mode Control for a Nonlinear Teleoperation System with Actuator Dynamics	2023	Journal of Automation, Mobile Robotics and Intelligent Sys
(Braun & Goldfarb, 2009)	A Control Approach for Actuated Dynamic Walking in Biped Robots	2009	IEEE Transactions on Robotics
(Song & Chang, 2017)	Managing impressions online: Microblogs and the state media's adaptation of online logics in China	2017	Journalism
(Zhang et al., 2026b)	From Noise to Control: Parameterized Diffusion Policies	2026	arXiv
(Shiller & Dubowsky, 2005)	On the optimal control of robotic manipulators with actuator and end-effector constraints	2005	openalex
(Falugi et al., 2010)	Multi-model predictive control based on LMI: from the adaptation of the state-space model to the analytic description	2010	International Journal of Control
(Ueda et al., 2007)	Broadcast Feedback of Stochastic Cellular Actuators Inspired by Biological Muscle Control	2007	The International Journal of Robotics Research
(Yonezawa et al., 2021)	Experimental Verification of Model-Free Vibration Control Technique Based on a Virtual Controlled Object Considering A	2021	Volume 7A: Dynamics, Vibration, and Control
(Bicego et al., 2020)	Nonlinear Model Predictive Control with Enhanced Actuator Model for Multi-Rotor Aerial Vehicles with Generic Designs	2020	Journal of Intelligent & Robotic Systems
(Kong et al., 2026)	Generative AI Against Poaching: Latent Composite Flow Matching for Poaching Prediction	2026	Proceedings of the AAAI Conference on Artificial Intelligence
(Balachandran et al., 2026)	Comparative Analysis of Intelligent PID Tuning Methods for Position Tracking in Mismatched Electro-Hydraulic Actuator	2026	International Journal of Robotics and Control Systems

Table 8: Local prior-work pressure map. The entries are used as threat coverage, not as an exhaustive manual survey. (continued)

Cite	Prior-work pressure	Year	Source
(Pasupuleti, 2026)	Neuromorphic Quantum Mechatronics for Soft Robotics, Assistive Gerontechnology, and Human-Machine Cognitive Adaptation	2026	International Journal of Academic and Industrial Research
(Pool, 2023)	Supplemental Material for Latent State-Trait and Latent Growth Curve Modeling of Inhibitory Control	2023	Journal of Experimental Psychology: General
(Wen & Tao, 2020)	Adaptive State Tracking Control with Actuator Nonlinearities and Failures	2020	International Conference of Control, Dynamic Systems, and
(Chappell et al., 2012)	Slat Noise Feedback Control with a Dielectric Barrier Discharge Plasma Actuator	2012	6th AIAA Flow Control Conference
(Rsetam et al., 2019)	Cascaded-Extended-State-Observer-Based Sliding-Mode Control for Underactuated Flexible Joint Robot	2019	IEEE Transactions on Industrial Electronics
(Wang et al., 2022)	Design and Dynamic Locomotion Control of Quadruped Robot with Perception-Less Terrain Adaptation	2022	Cyborg and Bionic Systems
(Ahmad, 2023)	Modeling and Hybrid PSO-WOA-Based Intelligent PID and State-Feedback Control for Ball and Beam Systems	2023	IEEE Access
(Chakrabartty et al., 2013)	Noise-Shaping Gradient Descent-Based Online Adaptation Algorithms for Digital Calibration of Analog Circuits	2013	IEEE Transactions on Neural Networks and Learning Systems
(Ghani et al., 2022)	Improved Third Order PID Sliding Mode Controller for Electrohydraulic Actuator Tracking Control	2022	Journal of Robotics and Control (JRC)
(Parthasarathy, 2022)	Random Feedback Control, Group Invariants and Pattern Classification, Quantum Mechanics in Levy Noise, State Observer	2022	Electromagnetics, Control and Robotics
(Qian et al., 2022)	Adaptive Oscillator-Based Assistive Torque Control for Gait Asymmetry Correction With a nSEA-Driven Hip Exoskeleton	2022	IEEE Transactions on Neural Systems and Rehabilitation Eng
(Hagh et al., 2021)	Active Fault-Tolerant Control Design for Actuator Fault Mitigation in Robotic Manipulators	2021	IEEE Access
(Phan et al., 2024)	Adaptive neural tracking control for flexible joint robot including hydraulic actuator dynamics with disturbance obser	2024	International Journal of Robust and Nonlinear Control
(Durand & Christofides, 2018)	Economic Model Predictive Control: Handling Valve Actuator Dynamics and Process Equipment Considerations	2018	Foundations and Trends in Systems and Control
(Perez et al., 2018)	Efficient transfer learning and online adaptation with latent variable models for continuous control	2018	arXiv
(Ye & Aghili, 2018)	Robot Manipulators Control with Parameters Adaptation and State Estimation	2018	2018 3rd International Conference on Advanced Robotics and
(Figwer & Michalczyk, 2018)	On Initialization of Adaptation in Active Noise Control	2018	2018 23rd International Conference on Methods & Models in
(Kong et al., 2019)	Asymmetric Bounded Neural Control for an Uncertain Robot by State Feedback and Output Feedback	2019	IEEE Transactions on Systems Man and Cybernetics Systems
(Wang et al., 2009)	Neural-Network-Based Terminal Sliding-Mode Control of Robotic Manipulators Including Actuator Dynamics	2009	IEEE Transactions on Industrial Electronics
(Zhao et al., 2023)	Design and Experiments of Electro-Hydrostatic Actuator for Wheel-Legged Robot with Fast Force Control Response	2023	Machines
(Yang et al., 2021)	Adaptive latent state modeling of brain network dynamics with real-time learning rate optimization	2021	Journal of Neural Engineering
(Sabelhaus et al., 2024)	Safe Supervisory Control of Soft Robot Actuators	2024	Soft Robotics
(Manurung et al., 2010)	Speed adaptation control of a small-sized treadmill with state feedback controller	2010	2010 3rd IEEE RAS & EMBS International Conference on Biom
(Michalczyk et al., 2016)	Adaptation in active noise control - A simulation case study	2016	2016 21st International Conference on Methods and Models i
(Zhang et al., 2026a)	Event-triggered model predictive control of positive systems with random actuator saturation		crossref
(Hu et al., 2026)	A novel proportional-integral-derivative -type iterative learning control strategy based on 2D model predictive iterat	2026	The Canadian Journal of Chemical Engineering
(Deshpande & Zhang, 2021)	Fault-Tolerant Model Predictive Control of a Fixed-Wing UAV with Actuator Fault Estimation	2021	Guidance, Navigation and Control
(Carlos et al., 2018)	An Analysis of Policies from Stochastic Linear Quadratic Gaussian in Robotics Problems with State- and Control-Depende	2018	Journal of Intelligent & Robotic Systems
(Lee & Park, 2023)	Circular active noise barrier using theoretical control filter considering interaction between speaker and barrier	2023	Scientific Reports

11 REPRODUCIBILITY

All tables and figures are regenerated by running `python src\run_experiment.py` from the repository root. The script streams raw CSVs and aggregates seed-level statistics. The canonical local PDF is `C:/Users/wangz/Downloads/86.pdf`; no Desktop copy is part of the artifact contract.

Table 9: Validated row counts for the v5 evidence package.

Artifact	Rows
rollouts.csv	199680
dataset_summary.csv	15360
raw_seed_metrics.csv	1040
metrics.csv	1352
pairwise_stats.csv	768
hard_aggregate_seed_metrics.csv	130
hard_aggregate_metrics.csv	169
hard_aggregate_pairwise_stats.csv	96
ablation_rollouts.csv	33600
ablation_seed_metrics.csv	200
ablation_metrics.csv	20
stress_sweep_raw.csv	302400
stress_sweep_seed_metrics.csv	2520
stress_sweep.csv	252
fixed_risk_raw.csv	69120
fixed_risk_seed_metrics.csv	480
fixed_risk_metrics.csv	48
fixed_risk_pairwise.csv	200
negative_cases.csv	24

12 DISCUSSION

The result is not pretty, but it is useful. The belief-state policy improves hard-aggregate success, and the stress sweep suggests the representation is not pointless. But v5 increases tracking error and safety violation relative to v4, and the fixed-risk gate exposes that its risk score cannot certify useful deployment at a strict budget. The fastest path to revival is not another synthetic table. The paper would need hardware evidence or an accepted high-fidelity simulator benchmark, learned policies rather than deterministic proxy components, and direct comparisons against modern fault observers, adaptive MPC, safe RL, and robust residual-control baselines. It would also need a risk calibration procedure that retains nonzero coverage under a meaningful safety budget.

13 CONCLUSION

Actuator asymmetry as policy state remains a compelling research direction. This expanded audit makes the paper more honest and much more rigorous, but it does not make it submission-ready. The correct terminal action is **KILL/ARCHIVE** until external robot or high-fidelity evidence and a mechanism that beats strong baselines are available.

A APPENDIX: FULL MAIN EVIDENCE

This appendix reports every split-method aggregate from the frozen main protocol. The purpose is to make the negative decision auditable: the hard aggregate is not hiding a strong positive result on another predefined split.

Table 10: Full split-by-method main metrics. This appendix table prevents cherry-picking by exposing every predefined split and method.

Split	Method	Success	Track.	Safety	Asym.	Energy	Utility
Nominal	Nominal	0.169 $\pm$ 0.015	0.625	0.172	0.131	0.766	-0.299
Nominal	Domain rand.	0.144 $\pm$ 0.015	0.641	0.191	0.120	0.731	-0.336
Nominal	Scalar cal.	0.154 $\pm$ 0.016	0.633	0.180	0.051	0.749	-0.307
Nominal	Online ID	0.165 $\pm$ 0.013	0.634	0.186	0.037	0.754	-0.295
Nominal	Adaptive MPC	0.133 $\pm$ 0.015	0.650	0.205	0.033	0.728	-0.341
Nominal	Robust tube	0.122 $\pm$ 0.015	0.667	0.229	0.065	0.675	-0.375
Nominal	CBF-FTC MPC	0.122 $\pm$ 0.014	0.682	0.264	0.040	0.643	-0.393
Nominal	UKF observer	0.148 $\pm$ 0.019	0.642	0.197	0.033	0.744	-0.319
Nominal	RNN state proxy	0.145 $\pm$ 0.014	0.639	0.195	0.033	0.755	-0.321
Nominal	L1 adaptive	0.121 $\pm$ 0.011	0.674	0.242	0.031	0.672	-0.379
Nominal	Latent state v4	0.150 $\pm$ 0.013	0.642	0.200	0.030	0.758	-0.320
Nominal	Belief state v5	0.158 $\pm$ 0.013	0.654	0.217	0.030	0.733	-0.324
Nominal	Oracle state	0.187 $\pm$ 0.018	0.656	0.238	0.033	0.734	-0.308
Gain shift	Nominal	0.118 $\pm$ 0.011	0.660	0.220	0.739	0.786	-0.506
Gain shift	Domain rand.	0.120 $\pm$ 0.013	0.672	0.240	0.679	0.744	-0.504
Gain shift	Scalar cal.	0.119 $\pm$ 0.010	0.661	0.214	0.295	0.810	-0.420
Gain shift	Online ID	0.131 $\pm$ 0.012	0.662	0.223	0.188	0.850	-0.394
Gain shift	Adaptive MPC	0.125 $\pm$ 0.012	0.672	0.234	0.170	0.814	-0.405
Gain shift	Robust tube	0.102 $\pm$ 0.011	0.694	0.267	0.378	0.704	-0.485
Gain shift	CBF-FTC MPC	0.119 $\pm$ 0.014	0.708	0.299	0.222	0.690	-0.461
Gain shift	UKF observer	0.122 $\pm$ 0.015	0.666	0.231	0.143	0.842	-0.399
Gain shift	RNN state proxy	0.139 $\pm$ 0.009	0.661	0.229	0.148	0.854	-0.382
Gain shift	L1 adaptive	0.096 $\pm$ 0.010	0.699	0.279	0.132	0.749	-0.458
Gain shift	Latent state v4	0.153 $\pm$ 0.012	0.662	0.230	0.118	0.858	-0.364
Gain shift	Belief state v5	0.141 $\pm$ 0.013	0.678	0.253	0.116	0.826	-0.393
Gain shift	Oracle state	0.197 $\pm$ 0.014	0.673	0.259	0.102	0.855	-0.336
Deadzone/sat.	Nominal	0.046 $\pm$ 0.010	0.721	0.295	0.292	0.879	-0.603
Deadzone/sat.	Domain rand.	0.049 $\pm$ 0.008	0.740	0.326	0.268	0.832	-0.619
Deadzone/sat.	Scalar cal.	0.049 $\pm$ 0.010	0.726	0.308	0.144	0.912	-0.577
Deadzone/sat.	Online ID	0.097 $\pm$ 0.014	0.693	0.265	0.193	1.068	-0.490
Deadzone/sat.	Adaptive MPC	0.080 $\pm$ 0.014	0.711	0.294	0.177	1.031	-0.527
Deadzone/sat.	Robust tube	0.057 $\pm$ 0.015	0.756	0.357	0.186	0.831	-0.605
Deadzone/sat.	CBF-FTC MPC	0.058 $\pm$ 0.012	0.760	0.377	0.181	0.871	-0.605
Deadzone/sat.	UKF observer	0.124 $\pm$ 0.011	0.684	0.262	0.210	1.133	-0.461
Deadzone/sat.	RNN state proxy	0.121 $\pm$ 0.015	0.681	0.259	0.205	1.142	-0.462
Deadzone/sat.	L1 adaptive	0.090 $\pm$ 0.013	0.727	0.323	0.195	0.995	-0.535
Deadzone/sat.	Latent state v4	0.146 $\pm$ 0.013	0.670	0.257	0.191	1.198	-0.433
Deadzone/sat.	Belief state v5	0.148 $\pm$ 0.011	0.684	0.272	0.191	1.161	-0.440
Deadzone/sat.	Oracle state	0.198 $\pm$ 0.019	0.664	0.264	0.195	1.261	-0.383
Thermal drift	Nominal	0.138 $\pm$ 0.012	0.648	0.209	0.427	0.795	-0.422
Thermal drift	Domain rand.	0.116 $\pm$ 0.010	0.665	0.230	0.417	0.757	-0.458
Thermal drift	Scalar cal.	0.135 $\pm$ 0.011	0.657	0.219	0.343	0.791	-0.417
Thermal drift	Online ID	0.117 $\pm$ 0.015	0.658	0.225	0.252	0.819	-0.420
Thermal drift	Adaptive MPC	0.124 $\pm$ 0.010	0.667	0.241	0.264	0.787	-0.425
Thermal drift	Robust tube	0.117 $\pm$ 0.012	0.690	0.271	0.340	0.706	-0.470
Thermal drift	CBF-FTC MPC	0.103 $\pm$ 0.012	0.708	0.304	0.284	0.683	-0.494
Thermal drift	UKF observer	0.134 $\pm$ 0.019	0.661	0.233	0.227	0.828	-0.403
Thermal drift	RNN state proxy	0.158 $\pm$ 0.020	0.657	0.227	0.235	0.833	-0.375
Thermal drift	L1 adaptive	0.120 $\pm$ 0.012	0.693	0.278	0.240	0.733	-0.450
Thermal drift	Latent state v4	0.148 $\pm$ 0.012	0.655	0.229	0.225	0.850	-0.386
Thermal drift	Belief state v5	0.171 $\pm$ 0.010	0.669	0.254	0.221	0.823	-0.379
Thermal drift	Oracle state	0.193 $\pm$ 0.011	0.668	0.263	0.193	0.862	-0.358
Latency/hyst.	Nominal	0.114 $\pm$ 0.021	0.681	0.254	0.361	0.830	-0.476
Latency/hyst.	Domain rand.	0.105 $\pm$ 0.014	0.697	0.276	0.332	0.784	-0.497
Latency/hyst.	Scalar cal.	0.123 $\pm$ 0.015	0.687	0.266	0.169	0.825	-0.440
Latency/hyst.	Online ID	0.132 $\pm$ 0.016	0.683	0.269	0.146	0.867	-0.424
Latency/hyst.	Adaptive MPC	0.107 $\pm$ 0.012	0.694	0.277	0.138	0.837	-0.455
Latency/hyst.	Robust tube	0.096 $\pm$ 0.019	0.716	0.308	0.206	0.732	-0.501
Latency/hyst.	CBF-FTC MPC	0.096 $\pm$ 0.022	0.734	0.344	0.158	0.719	-0.515
Latency/hyst.	UKF observer	0.132 $\pm$ 0.018	0.688	0.281	0.138	0.888	-0.431
Latency/hyst.	RNN state proxy	0.140 $\pm$ 0.019	0.678	0.267	0.136	0.896	-0.412
Latency/hyst.	L1 adaptive	0.102 $\pm$ 0.014	0.716	0.317	0.132	0.778	-0.483
Latency/hyst.	Latent state v4	0.136 $\pm$ 0.020	0.678	0.266	0.128	0.929	-0.416
Latency/hyst.	Belief state v5	0.156 $\pm$ 0.018	0.688	0.286	0.129	0.893	-0.408
Latency/hyst.	Oracle state	0.196 $\pm$ 0.017	0.685	0.293	0.131	0.937	-0.373
Fault	Nominal	0.096 $\pm$ 0.014	0.673	0.243	0.433	0.812	-0.499
Fault	Domain rand.	0.087 $\pm$ 0.006	0.690	0.265	0.400	0.769	-0.519
Fault	Scalar cal.	0.104 $\pm$ 0.014	0.686	0.264	0.216	0.822	-0.469
Fault	Online ID	0.132 $\pm$ 0.018	0.675	0.254	0.198	0.880	-0.423
Fault	Adaptive MPC	0.103 $\pm$ 0.017	0.691	0.273	0.188	0.851	-0.467
Fault	Robust tube	0.087 $\pm$ 0.010	0.719	0.315	0.260	0.735	-0.527
Fault	CBF-FTC MPC	0.090 $\pm$ 0.014	0.730	0.340	0.208	0.726	-0.527
Fault	UKF observer	0.129 $\pm$ 0.013	0.679	0.265	0.196	0.899	-0.433
Fault	RNN state proxy	0.128 $\pm$ 0.015	0.673	0.260	0.191	0.910	-0.429
Fault	L1 adaptive	0.107 $\pm$ 0.013	0.708	0.300	0.187	0.797	-0.478
Fault	Latent state v4	0.134 $\pm$ 0.010	0.671	0.255	0.185	0.945	-0.420
Fault	Belief state v5	0.165 $\pm$ 0.017	0.681	0.274	0.187	0.914	-0.403

Table 10: Full split-by-method main metrics. This appendix table prevents cherry-picking by exposing every predefined split and method. (continued)

Split	Method	Success	Track.	Safety	Asym.	Energy	Utility
Fault	Oracle state	0.194 $\pm$ 0.018	0.677	0.279	0.191	0.961	-0.376
Payload/contact	Nominal	0.107 $\pm$ 0.021	0.693	0.276	0.435	0.842	-0.521
Payload/contact	Domain rand.	0.089 $\pm$ 0.014	0.704	0.293	0.404	0.787	-0.543
Payload/contact	Scalar cal.	0.087 $\pm$ 0.012	0.698	0.278	0.238	0.846	-0.506
Payload/contact	Online ID	0.104 $\pm$ 0.016	0.692	0.277	0.219	0.918	-0.478
Payload/contact	Adaptive MPC	0.096 $\pm$ 0.014	0.700	0.289	0.209	0.884	-0.492
Payload/contact	Robust tube	0.077 $\pm$ 0.009	0.733	0.339	0.277	0.751	-0.562
Payload/contact	CBF-FTC MPC	0.096 $\pm$ 0.013	0.739	0.355	0.230	0.746	-0.538
Payload/contact	UKF observer	0.123 $\pm$ 0.017	0.692	0.290	0.212	0.945	-0.464
Payload/contact	RNN state proxy	0.116 $\pm$ 0.016	0.682	0.276	0.211	0.958	-0.461
Payload/contact	L1 adaptive	0.092 $\pm$ 0.014	0.721	0.325	0.209	0.829	-0.518
Payload/contact	Latent state v4	0.121 $\pm$ 0.008	0.676	0.265	0.202	0.990	-0.448
Payload/contact	Belief state v5	0.154 $\pm$ 0.019	0.689	0.290	0.206	0.956	-0.432
Payload/contact	Oracle state	0.192 $\pm$ 0.017	0.678	0.291	0.203	1.013	-0.391
Combined	Nominal	0.066 $\pm$ 0.015	0.736	0.346	0.748	0.895	-0.683
Combined	Domain rand.	0.059 $\pm$ 0.008	0.759	0.379	0.689	0.842	-0.704
Combined	Scalar cal.	0.072 $\pm$ 0.020	0.744	0.354	0.392	0.937	-0.623
Combined	Online ID	0.089 $\pm$ 0.006	0.729	0.342	0.388	1.055	-0.586
Combined	Adaptive MPC	0.089 $\pm$ 0.014	0.747	0.367	0.364	1.007	-0.601
Combined	Robust tube	0.061 $\pm$ 0.009	0.775	0.400	0.474	0.811	-0.672
Combined	CBF-FTC MPC	0.070 $\pm$ 0.014	0.789	0.430	0.403	0.830	-0.666
Combined	UKF observer	0.089 $\pm$ 0.014	0.730	0.348	0.378	1.089	-0.587
Combined	RNN state proxy	0.111 $\pm$ 0.009	0.723	0.336	0.375	1.105	-0.557
Combined	L1 adaptive	0.083 $\pm$ 0.012	0.763	0.391	0.369	0.941	-0.618
Combined	Latent state v4	0.117 $\pm$ 0.018	0.714	0.329	0.342	1.145	-0.540
Combined	Belief state v5	0.119 $\pm$ 0.013	0.731	0.350	0.342	1.096	-0.553
Combined	Oracle state	0.169 $\pm$ 0.014	0.716	0.348	0.337	1.183	-0.497

B APPENDIX: HARD-AGGREGATE SEED UNITS

The following seed means are the statistical units used for hard-aggregate paired tests. They show that the mixed conclusion is not a formatting artifact; the safety and tracking failures appear across seed-level aggregates.

Table 11: Hard-aggregate seed means. These rows are the units used by paired tests.

Method	Seed	Success	Tracking	Safety	Asym. err.	Utility
Adaptive MPC	0	0.090	0.718	0.316	0.268	-0.534
Adaptive MPC	1	0.109	0.712	0.308	0.253	-0.505
Adaptive MPC	2	0.099	0.708	0.311	0.251	-0.515
Adaptive MPC	3	0.097	0.717	0.314	0.249	-0.522
Adaptive MPC	4	0.108	0.712	0.320	0.250	-0.514
Adaptive MPC	5	0.097	0.705	0.290	0.257	-0.504
Adaptive MPC	6	0.087	0.711	0.303	0.255	-0.525
Adaptive MPC	7	0.083	0.715	0.313	0.247	-0.535
Adaptive MPC	8	0.090	0.722	0.321	0.255	-0.537
Adaptive MPC	9	0.099	0.706	0.302	0.252	-0.509
CBF-FTC MPC	0	0.078	0.772	0.405	0.291	-0.612
CBF-FTC MPC	1	0.076	0.747	0.370	0.278	-0.581
CBF-FTC MPC	2	0.095	0.741	0.365	0.276	-0.555
CBF-FTC MPC	3	0.102	0.747	0.373	0.282	-0.557
CBF-FTC MPC	4	0.082	0.772	0.387	0.273	-0.596
CBF-FTC MPC	5	0.094	0.744	0.365	0.284	-0.561
CBF-FTC MPC	6	0.101	0.747	0.369	0.282	-0.556
CBF-FTC MPC	7	0.082	0.757	0.385	0.276	-0.588
CBF-FTC MPC	8	0.071	0.755	0.380	0.283	-0.595
CBF-FTC MPC	9	0.073	0.743	0.350	0.279	-0.572
Domain rand.	0	0.090	0.723	0.318	0.510	-0.584
Domain rand.	1	0.080	0.714	0.306	0.501	-0.583
Domain rand.	2	0.078	0.720	0.320	0.500	-0.595
Domain rand.	3	0.061	0.718	0.308	0.489	-0.603
Domain rand.	4	0.073	0.728	0.332	0.485	-0.609
Domain rand.	5	0.087	0.718	0.312	0.506	-0.582
Domain rand.	6	0.071	0.711	0.294	0.499	-0.584
Domain rand.	7	0.080	0.719	0.323	0.495	-0.594
Domain rand.	8	0.073	0.719	0.312	0.493	-0.594
Domain rand.	9	0.090	0.706	0.296	0.499	-0.561
Belief state v5	0	0.108	0.716	0.326	0.255	-0.521
Belief state v5	1	0.165	0.692	0.295	0.243	-0.434
Belief state v5	2	0.161	0.694	0.304	0.240	-0.442
Belief state v5	3	0.144	0.703	0.306	0.244	-0.466
Belief state v5	4	0.134	0.705	0.313	0.242	-0.482
Belief state v5	5	0.135	0.692	0.289	0.252	-0.461
Belief state v5	6	0.174	0.692	0.282	0.248	-0.419
Belief state v5	7	0.149	0.709	0.327	0.232	-0.473
Belief state v5	8	0.137	0.709	0.315	0.248	-0.481

Table 11: Hard-aggregate seed means. These rows are the units used by paired tests. (continued)

Method	Seed	Success	Tracking	Safety	Asym. err.	Utility
Belief state v5	9	0.149	0.688	0.289	0.247	-0.445
Latent state v4	0	0.134	0.696	0.290	0.249	-0.469
Latent state v4	1	0.123	0.676	0.273	0.241	-0.460
Latent state v4	2	0.115	0.686	0.287	0.240	-0.480
Latent state v4	3	0.149	0.680	0.270	0.245	-0.435
Latent state v4	4	0.118	0.707	0.316	0.235	-0.502
Latent state v4	5	0.135	0.679	0.269	0.250	-0.449
Latent state v4	6	0.132	0.675	0.261	0.246	-0.445
Latent state v4	7	0.094	0.696	0.293	0.230	-0.508
Latent state v4	8	0.127	0.685	0.284	0.244	-0.467
Latent state v4	9	0.115	0.688	0.284	0.247	-0.481
Nominal	0	0.076	0.717	0.307	0.552	-0.602
Nominal	1	0.123	0.690	0.274	0.541	-0.522
Nominal	2	0.109	0.688	0.282	0.541	-0.539
Nominal	3	0.071	0.700	0.280	0.530	-0.579
Nominal	4	0.075	0.710	0.308	0.525	-0.597
Nominal	5	0.071	0.705	0.292	0.547	-0.592
Nominal	6	0.095	0.697	0.280	0.540	-0.555
Nominal	7	0.078	0.700	0.291	0.536	-0.580
Nominal	8	0.106	0.698	0.288	0.534	-0.548
Nominal	9	0.090	0.698	0.281	0.540	-0.561
Online ID	0	0.106	0.706	0.298	0.278	-0.507
Online ID	1	0.109	0.691	0.287	0.268	-0.487
Online ID	2	0.101	0.697	0.300	0.266	-0.506
Online ID	3	0.108	0.698	0.283	0.266	-0.492
Online ID	4	0.099	0.713	0.312	0.263	-0.523
Online ID	5	0.128	0.691	0.280	0.271	-0.466
Online ID	6	0.090	0.692	0.272	0.269	-0.500
Online ID	7	0.128	0.703	0.300	0.260	-0.481
Online ID	8	0.092	0.700	0.295	0.270	-0.515
Online ID	9	0.120	0.695	0.283	0.271	-0.478
Oracle state	0	0.168	0.707	0.328	0.252	-0.459
Oracle state	1	0.177	0.685	0.301	0.241	-0.423
Oracle state	2	0.193	0.689	0.308	0.237	-0.412
Oracle state	3	0.196	0.684	0.289	0.243	-0.398
Oracle state	4	0.168	0.705	0.336	0.240	-0.461
Oracle state	5	0.177	0.690	0.306	0.254	-0.430
Oracle state	6	0.227	0.676	0.280	0.245	-0.358
Oracle state	7	0.189	0.694	0.322	0.235	-0.425
Oracle state	8	0.186	0.690	0.305	0.248	-0.420
Oracle state	9	0.168	0.683	0.288	0.242	-0.426
RNN state proxy	0	0.130	0.697	0.296	0.269	-0.477
RNN state proxy	1	0.116	0.683	0.280	0.259	-0.475
RNN state proxy	2	0.116	0.690	0.298	0.258	-0.487
RNN state proxy	3	0.123	0.692	0.287	0.260	-0.475
RNN state proxy	4	0.108	0.703	0.311	0.254	-0.508
RNN state proxy	5	0.111	0.697	0.291	0.267	-0.494
RNN state proxy	6	0.122	0.696	0.290	0.263	-0.481
RNN state proxy	7	0.125	0.696	0.298	0.243	-0.479
RNN state proxy	8	0.095	0.697	0.291	0.264	-0.508
RNN state proxy	9	0.137	0.674	0.263	0.256	-0.439
L1 adaptive	0	0.078	0.736	0.342	0.266	-0.561
L1 adaptive	1	0.108	0.724	0.322	0.256	-0.512
L1 adaptive	2	0.108	0.727	0.337	0.250	-0.520
L1 adaptive	3	0.094	0.723	0.332	0.252	-0.531
L1 adaptive	4	0.094	0.739	0.355	0.253	-0.551
L1 adaptive	5	0.099	0.727	0.329	0.260	-0.527
L1 adaptive	6	0.101	0.736	0.342	0.257	-0.537
L1 adaptive	7	0.082	0.735	0.355	0.247	-0.560
L1 adaptive	8	0.090	0.731	0.335	0.251	-0.540
L1 adaptive	9	0.089	0.728	0.339	0.258	-0.542
Robust tube	0	0.062	0.762	0.376	0.349	-0.624
Robust tube	1	0.082	0.730	0.328	0.338	-0.562
Robust tube	2	0.071	0.732	0.344	0.336	-0.582
Robust tube	3	0.069	0.748	0.352	0.329	-0.594
Robust tube	4	0.078	0.751	0.368	0.327	-0.597
Robust tube	5	0.066	0.742	0.348	0.345	-0.597
Robust tube	6	0.080	0.738	0.347	0.338	-0.577
Robust tube	7	0.087	0.742	0.356	0.332	-0.576
Robust tube	8	0.073	0.740	0.349	0.337	-0.587
Robust tube	9	0.082	0.736	0.345	0.338	-0.573
Scalar cal.	0	0.092	0.718	0.308	0.292	-0.540
Scalar cal.	1	0.118	0.702	0.287	0.285	-0.493
Scalar cal.	2	0.109	0.701	0.297	0.285	-0.505
Scalar cal.	3	0.075	0.707	0.293	0.276	-0.540
Scalar cal.	4	0.085	0.716	0.314	0.273	-0.546
Scalar cal.	5	0.080	0.708	0.305	0.288	-0.545
Scalar cal.	6	0.071	0.711	0.287	0.282	-0.544
Scalar cal.	7	0.089	0.711	0.311	0.276	-0.538
Scalar cal.	8	0.071	0.714	0.299	0.281	-0.551
Scalar cal.	9	0.085	0.704	0.285	0.281	-0.525
UKF observer	0	0.111	0.713	0.313	0.273	-0.512
UKF observer	1	0.113	0.703	0.307	0.260	-0.500
UKF observer	2	0.120	0.696	0.304	0.256	-0.487
UKF observer	3	0.116	0.700	0.305	0.262	-0.494
UKF observer	4	0.104	0.707	0.318	0.254	-0.517
UKF observer	5	0.108	0.695	0.294	0.272	-0.497
UKF observer	6	0.108	0.691	0.276	0.263	-0.483
UKF observer	7	0.111	0.703	0.298	0.251	-0.495

Table 11: Hard-aggregate seed means. These rows are the units used by paired tests. (continued)

Method	Seed	Success	Tracking	Safety	Asym. err.	Utility
UKF observer	8	0.128	0.706	0.309	0.265	-0.487
UKF observer	9	0.120	0.693	0.286	0.263	-0.477

C APPENDIX: COMPLETE PAIRED TESTS

The table below gives every hard-aggregate paired difference for v5 minus each reference method across the predefined paired metrics. Primary success succeeds against v4, but tracking and safety fail against v4.

Table 12: All hard-aggregate paired differences for belief state v5 minus each reference method.

Reference	Metric	Mean diff	CI95	Lower95	Upper95
Adaptive MPC	task_success	0.050	0.012	0.037	0.062
Adaptive MPC	tracking_error	-0.012	0.004	-0.016	-0.009
Adaptive MPC	safety_violation	-0.005	0.006	-0.011	0.002
Adaptive MPC	asymmetry_error	-0.009	0.002	-0.011	-0.007
Adaptive MPC	actuator_energy	0.075	0.003	0.072	0.078
Adaptive MPC	adaptation_regret	-0.003	0.001	-0.005	-0.002
Adaptive MPC	constraint_margin	0.012	0.004	0.009	0.016
Adaptive MPC	robust_utility	0.058	0.016	0.042	0.073
CBF-FTC MPC	task_success	0.060	0.011	0.049	0.072
CBF-FTC MPC	tracking_error	-0.052	0.004	-0.057	-0.048
CBF-FTC MPC	safety_violation	-0.070	0.006	-0.076	-0.064
CBF-FTC MPC	asymmetry_error	-0.035	0.002	-0.038	-0.033
CBF-FTC MPC	actuator_energy	0.221	0.004	0.217	0.225
CBF-FTC MPC	adaptation_regret	-0.015	0.001	-0.016	-0.014
CBF-FTC MPC	constraint_margin	0.052	0.004	0.048	0.057
CBF-FTC MPC	robust_utility	0.115	0.012	0.103	0.126
Domain rand.	task_success	0.067	0.015	0.053	0.082
Domain rand.	tracking_error	-0.017	0.004	-0.022	-0.013
Domain rand.	safety_violation	-0.007	0.006	-0.014	-0.001
Domain rand.	asymmetry_error	-0.253	0.004	-0.257	-0.248
Domain rand.	actuator_energy	0.189	0.003	0.186	0.192
Domain rand.	adaptation_regret	-0.005	0.001	-0.006	-0.004
Domain rand.	constraint_margin	0.017	0.004	0.013	0.022
Domain rand.	robust_utility	0.126	0.017	0.109	0.144
Latent state v4	task_success	0.022	0.017	0.005	0.038
Latent state v4	tracking_error	0.013	0.006	0.008	0.019
Latent state v4	safety_violation	0.022	0.008	0.014	0.030
Latent state v4	asymmetry_error	0.002	0.001	0.001	0.004
Latent state v4	actuator_energy	-0.038	0.004	-0.042	-0.034
Latent state v4	adaptation_regret	0.004	0.002	0.002	0.005
Latent state v4	constraint_margin	-0.013	0.006	-0.019	-0.008
Latent state v4	robust_utility	0.007	0.020	-0.013	0.027
Nominal	task_success	0.056	0.010	0.046	0.066
Nominal	tracking_error	-0.000	0.005	-0.005	0.005
Nominal	safety_violation	0.016	0.008	0.009	0.024
Nominal	asymmetry_error	-0.294	0.004	-0.298	-0.289
Nominal	actuator_energy	0.139	0.007	0.132	0.146
Nominal	adaptation_regret	-0.000	0.001	-0.001	0.001
Nominal	constraint_margin	0.001	0.005	-0.004	0.005
Nominal	robust_utility	0.105	0.014	0.092	0.119
Online ID	task_success	0.037	0.015	0.022	0.053
Online ID	tracking_error	0.002	0.004	-0.002	0.006
Online ID	safety_violation	0.014	0.006	0.007	0.020
Online ID	asymmetry_error	-0.023	0.002	-0.025	-0.022
Online ID	actuator_energy	0.038	0.004	0.033	0.042
Online ID	adaptation_regret	0.000	0.001	-0.001	0.002
Online ID	constraint_margin	-0.002	0.004	-0.006	0.002
Online ID	robust_utility	0.033	0.018	0.015	0.051
Oracle state	task_success	-0.039	0.010	-0.049	-0.030
Oracle state	tracking_error	0.010	0.005	0.005	0.015
Oracle state	safety_violation	-0.002	0.007	-0.009	0.006
Oracle state	asymmetry_error	0.001	0.001	-0.000	0.003
Oracle state	actuator_energy	-0.064	0.004	-0.068	-0.060
Oracle state	adaptation_regret	-0.066	0.002	-0.068	-0.065
Oracle state	constraint_margin	-0.010	0.005	-0.015	-0.005
Oracle state	robust_utility	-0.041	0.013	-0.054	-0.028
RNN state proxy	task_success	0.027	0.014	0.014	0.041
RNN state proxy	tracking_error	0.008	0.005	0.003	0.013
RNN state proxy	safety_violation	0.014	0.009	0.006	0.023
RNN state proxy	asymmetry_error	-0.014	0.002	-0.016	-0.013
RNN state proxy	actuator_energy	-0.002	0.005	-0.007	0.002
RNN state proxy	adaptation_regret	0.002	0.001	0.001	0.004
RNN state proxy	constraint_margin	-0.008	0.005	-0.013	-0.003
RNN state proxy	robust_utility	0.020	0.019	0.001	0.038
L1 adaptive	task_success	0.052	0.009	0.043	0.060

Table 12: All hard-aggregate paired differences for belief state v5 minus each reference method. (continued)

Reference	Metric	Mean diff	CI95	Lower95	Upper95
L1 adaptive	tracking_error	-0.030	0.005	-0.036	-0.025
L1 adaptive	safety_violation	-0.034	0.009	-0.043	-0.026
L1 adaptive	asymmetry_error	-0.010	0.002	-0.012	-0.008
L1 adaptive	actuator_energy	0.133	0.003	0.130	0.137
L1 adaptive	adaptation_regret	-0.009	0.001	-0.010	-0.007
L1 adaptive	constraint_margin	0.030	0.005	0.025	0.036
L1 adaptive	robust_utility	0.076	0.013	0.062	0.089
Robust tube	task_success	0.071	0.009	0.061	0.080
Robust tube	tracking_error	-0.042	0.004	-0.046	-0.038
Robust tube	safety_violation	-0.047	0.008	-0.054	-0.039
Robust tube	asymmetry_error	-0.092	0.003	-0.095	-0.089
Robust tube	actuator_energy	0.223	0.004	0.218	0.227
Robust tube	adaptation_regret	-0.012	0.001	-0.013	-0.011
Robust tube	constraint_margin	0.042	0.004	0.038	0.046
Robust tube	robust_utility	0.124	0.011	0.113	0.136
Scalar cal.	task_success	0.058	0.013	0.045	0.072
Scalar cal.	tracking_error	-0.009	0.004	-0.013	-0.005
Scalar cal.	safety_violation	0.006	0.007	-0.001	0.013
Scalar cal.	asymmetry_error	-0.037	0.003	-0.040	-0.034
Scalar cal.	actuator_energy	0.120	0.004	0.116	0.124
Scalar cal.	adaptation_regret	-0.003	0.001	-0.004	-0.001
Scalar cal.	constraint_margin	0.009	0.004	0.005	0.013
Scalar cal.	robust_utility	0.070	0.016	0.054	0.086
UKF observer	task_success	0.032	0.012	0.019	0.044
UKF observer	tracking_error	-0.000	0.003	-0.004	0.003
UKF observer	safety_violation	0.004	0.007	-0.003	0.011
UKF observer	asymmetry_error	-0.017	0.001	-0.018	-0.016
UKF observer	actuator_energy	0.011	0.003	0.008	0.014
UKF observer	adaptation_regret	-0.000	0.001	-0.001	0.001
UKF observer	constraint_margin	0.000	0.003	-0.003	0.004
UKF observer	robust_utility	0.032	0.014	0.018	0.047

D APPENDIX: COMPLETE STRESS SWEEP

The main text highlights combined stress because it is the most important deployment stressor, but all axes are reported here. This table includes gain asymmetry, deadzone/saturation, latency, thermal drift, intermittent fault, and combined stress.

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods.

Axis	Level	Method	Success	Track.	Safety	Asym.	Utility
gain_asymmetry	0.0	Domain rand.	0.064	0.746	0.353	0.297	-0.616
gain_asymmetry	0.0	Online ID	0.104	0.719	0.317	0.248	-0.526
gain_asymmetry	0.0	CBF-FTC MPC	0.055	0.770	0.398	0.264	-0.632
gain_asymmetry	0.0	UKF observer	0.115	0.714	0.321	0.260	-0.518
gain_asymmetry	0.0	L1 adaptive	0.092	0.746	0.360	0.263	-0.568
gain_asymmetry	0.0	Belief state v5	0.117	0.718	0.321	0.254	-0.518
gain_asymmetry	0.0	Oracle state	0.162	0.699	0.315	0.243	-0.463
gain_asymmetry	0.2	Domain rand.	0.054	0.745	0.351	0.363	-0.635
gain_asymmetry	0.2	Online ID	0.099	0.712	0.316	0.278	-0.532
gain_asymmetry	0.2	CBF-FTC MPC	0.083	0.779	0.415	0.288	-0.622
gain_asymmetry	0.2	UKF observer	0.098	0.708	0.308	0.278	-0.528
gain_asymmetry	0.2	L1 adaptive	0.102	0.744	0.361	0.282	-0.559
gain_asymmetry	0.2	Belief state v5	0.130	0.710	0.318	0.269	-0.503
gain_asymmetry	0.2	Oracle state	0.172	0.702	0.327	0.260	-0.464
gain_asymmetry	0.4	Domain rand.	0.074	0.748	0.358	0.484	-0.640
gain_asymmetry	0.4	Online ID	0.098	0.721	0.322	0.321	-0.550
gain_asymmetry	0.4	CBF-FTC MPC	0.074	0.784	0.421	0.333	-0.643
gain_asymmetry	0.4	UKF observer	0.124	0.715	0.322	0.316	-0.519
gain_asymmetry	0.4	L1 adaptive	0.091	0.756	0.378	0.315	-0.591
gain_asymmetry	0.4	Belief state v5	0.122	0.719	0.337	0.296	-0.529
gain_asymmetry	0.4	Oracle state	0.175	0.703	0.321	0.284	-0.462
gain_asymmetry	0.6	Domain rand.	0.069	0.750	0.360	0.634	-0.671
gain_asymmetry	0.6	Online ID	0.107	0.722	0.334	0.367	-0.555
gain_asymmetry	0.6	CBF-FTC MPC	0.074	0.774	0.403	0.378	-0.636
gain_asymmetry	0.6	UKF observer	0.093	0.723	0.333	0.359	-0.568
gain_asymmetry	0.6	L1 adaptive	0.095	0.756	0.384	0.351	-0.596
gain_asymmetry	0.6	Belief state v5	0.121	0.720	0.334	0.335	-0.535
gain_asymmetry	0.6	Oracle state	0.174	0.704	0.331	0.322	-0.475
gain_asymmetry	0.8	Domain rand.	0.062	0.755	0.367	0.845	-0.720
gain_asymmetry	0.8	Online ID	0.075	0.743	0.349	0.447	-0.621
gain_asymmetry	0.8	CBF-FTC MPC	0.072	0.786	0.417	0.461	-0.666
gain_asymmetry	0.8	UKF observer	0.083	0.740	0.352	0.432	-0.610
gain_asymmetry	0.8	L1 adaptive	0.063	0.773	0.405	0.416	-0.659
gain_asymmetry	0.8	Belief state v5	0.129	0.738	0.356	0.386	-0.555
gain_asymmetry	0.8	Oracle state	0.146	0.728	0.360	0.381	-0.540
gain_asymmetry	1.0	Domain rand.	0.069	0.766	0.389	1.023	-0.760

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods. (continued)

Axis	Level	Method	Success	Track.	Safety	Asym.	Utility
gain_asymmetry	1.0	Online ID	0.072	0.752	0.366	0.519	-0.652
gain_asymmetry	1.0	CBF-FTC MPC	0.063	0.803	0.438	0.540	-0.707
gain_asymmetry	1.0	UKF observer	0.075	0.756	0.375	0.489	-0.648
gain_asymmetry	1.0	L1 adaptive	0.057	0.789	0.421	0.475	-0.693
gain_asymmetry	1.0	Belief state v5	0.087	0.758	0.381	0.439	-0.630
gain_asymmetry	1.0	Oracle state	0.135	0.741	0.365	0.429	-0.568
deadzone_saturation	0.0	Domain rand.	0.117	0.696	0.297	0.702	-0.554
deadzone_saturation	0.0	Online ID	0.127	0.688	0.290	0.246	-0.457
deadzone_saturation	0.0	CBF-FTC MPC	0.102	0.739	0.365	0.289	-0.539
deadzone_saturation	0.0	UKF observer	0.127	0.698	0.305	0.195	-0.458
deadzone_saturation	0.0	L1 adaptive	0.105	0.719	0.335	0.201	-0.499
deadzone_saturation	0.0	Belief state v5	0.138	0.711	0.330	0.173	-0.462
deadzone_saturation	0.0	Oracle state	0.161	0.701	0.321	0.132	-0.422
deadzone_saturation	0.2	Domain rand.	0.098	0.716	0.317	0.706	-0.601
deadzone_saturation	0.2	Online ID	0.102	0.705	0.301	0.313	-0.515
deadzone_saturation	0.2	CBF-FTC MPC	0.092	0.760	0.386	0.343	-0.587
deadzone_saturation	0.2	UKF observer	0.102	0.708	0.318	0.282	-0.520
deadzone_saturation	0.2	L1 adaptive	0.078	0.743	0.361	0.278	-0.573
deadzone_saturation	0.2	Belief state v5	0.132	0.709	0.324	0.263	-0.490
deadzone_saturation	0.2	Oracle state	0.157	0.709	0.330	0.249	-0.468
deadzone_saturation	0.4	Domain rand.	0.062	0.743	0.345	0.695	-0.674
deadzone_saturation	0.4	Online ID	0.092	0.724	0.326	0.372	-0.566
deadzone_saturation	0.4	CBF-FTC MPC	0.094	0.773	0.400	0.390	-0.614
deadzone_saturation	0.4	UKF observer	0.100	0.724	0.339	0.355	-0.562
deadzone_saturation	0.4	L1 adaptive	0.089	0.756	0.380	0.352	-0.598
deadzone_saturation	0.4	Belief state v5	0.130	0.725	0.341	0.325	-0.528
deadzone_saturation	0.4	Oracle state	0.162	0.712	0.341	0.318	-0.492
deadzone_saturation	0.6	Domain rand.	0.049	0.769	0.387	0.695	-0.730
deadzone_saturation	0.6	Online ID	0.083	0.737	0.343	0.415	-0.603
deadzone_saturation	0.6	CBF-FTC MPC	0.056	0.795	0.430	0.417	-0.689
deadzone_saturation	0.6	UKF observer	0.091	0.734	0.357	0.407	-0.600
deadzone_saturation	0.6	L1 adaptive	0.087	0.771	0.396	0.395	-0.629
deadzone_saturation	0.6	Belief state v5	0.101	0.733	0.352	0.367	-0.581
deadzone_saturation	0.6	Oracle state	0.152	0.712	0.335	0.361	-0.514
deadzone_saturation	0.8	Domain rand.	0.046	0.783	0.405	0.689	-0.757
deadzone_saturation	0.8	Online ID	0.085	0.745	0.364	0.435	-0.627
deadzone_saturation	0.8	CBF-FTC MPC	0.062	0.810	0.450	0.439	-0.709
deadzone_saturation	0.8	UKF observer	0.073	0.752	0.373	0.428	-0.647
deadzone_saturation	0.8	L1 adaptive	0.067	0.787	0.406	0.421	-0.672
deadzone_saturation	0.8	Belief state v5	0.100	0.745	0.363	0.378	-0.603
deadzone_saturation	0.8	Oracle state	0.148	0.726	0.346	0.377	-0.542
deadzone_saturation	1.0	Domain rand.	0.037	0.811	0.444	0.697	-0.808
deadzone_saturation	1.0	Online ID	0.068	0.763	0.373	0.458	-0.667
deadzone_saturation	1.0	CBF-FTC MPC	0.047	0.831	0.480	0.454	-0.758
deadzone_saturation	1.0	UKF observer	0.081	0.757	0.378	0.448	-0.652
deadzone_saturation	1.0	L1 adaptive	0.062	0.802	0.432	0.439	-0.703
deadzone_saturation	1.0	Belief state v5	0.102	0.756	0.388	0.397	-0.627
deadzone_saturation	1.0	Oracle state	0.125	0.730	0.357	0.395	-0.582
latency	0.0	Domain rand.	0.063	0.729	0.333	0.709	-0.664
latency	0.0	Online ID	0.098	0.707	0.296	0.408	-0.543
latency	0.0	CBF-FTC MPC	0.090	0.760	0.380	0.418	-0.607
latency	0.0	UKF observer	0.119	0.693	0.301	0.394	-0.515
latency	0.0	L1 adaptive	0.077	0.737	0.347	0.389	-0.590
latency	0.0	Belief state v5	0.144	0.701	0.309	0.371	-0.495
latency	0.0	Oracle state	0.193	0.686	0.299	0.360	-0.436
latency	0.2	Domain rand.	0.061	0.751	0.354	0.698	-0.687
latency	0.2	Online ID	0.093	0.721	0.319	0.395	-0.566
latency	0.2	CBF-FTC MPC	0.069	0.778	0.403	0.411	-0.648
latency	0.2	UKF observer	0.107	0.713	0.325	0.388	-0.549
latency	0.2	L1 adaptive	0.079	0.758	0.379	0.383	-0.616
latency	0.2	Belief state v5	0.117	0.718	0.332	0.360	-0.541
latency	0.2	Oracle state	0.176	0.707	0.328	0.352	-0.477
latency	0.4	Domain rand.	0.063	0.751	0.357	0.705	-0.687
latency	0.4	Online ID	0.092	0.730	0.323	0.394	-0.573
latency	0.4	CBF-FTC MPC	0.069	0.794	0.432	0.412	-0.671
latency	0.4	UKF observer	0.107	0.724	0.339	0.383	-0.560
latency	0.4	L1 adaptive	0.064	0.778	0.398	0.375	-0.648
latency	0.4	Belief state v5	0.122	0.730	0.348	0.353	-0.550
latency	0.4	Oracle state	0.142	0.724	0.346	0.339	-0.527
latency	0.6	Domain rand.	0.069	0.749	0.361	0.706	-0.683
latency	0.6	Online ID	0.096	0.723	0.320	0.398	-0.565
latency	0.6	CBF-FTC MPC	0.064	0.793	0.429	0.413	-0.674
latency	0.6	UKF observer	0.089	0.729	0.344	0.390	-0.586
latency	0.6	L1 adaptive	0.079	0.768	0.397	0.376	-0.630
latency	0.6	Belief state v5	0.144	0.728	0.353	0.353	-0.530
latency	0.6	Oracle state	0.158	0.718	0.340	0.341	-0.506
latency	0.8	Domain rand.	0.070	0.757	0.368	0.701	-0.688
latency	0.8	Online ID	0.099	0.732	0.342	0.393	-0.578
latency	0.8	CBF-FTC MPC	0.083	0.790	0.432	0.405	-0.656
latency	0.8	UKF observer	0.096	0.731	0.343	0.384	-0.580
latency	0.8	L1 adaptive	0.084	0.767	0.391	0.366	-0.619
latency	0.8	Belief state v5	0.128	0.730	0.342	0.343	-0.540
latency	0.8	Oracle state	0.157	0.723	0.352	0.332	-0.516
latency	1.0	Domain rand.	0.058	0.767	0.382	0.699	-0.712
latency	1.0	Online ID	0.095	0.737	0.346	0.388	-0.585
latency	1.0	CBF-FTC MPC	0.065	0.801	0.439	0.398	-0.681
latency	1.0	UKF observer	0.091	0.743	0.362	0.373	-0.598
latency	1.0	L1 adaptive	0.087	0.773	0.399	0.358	-0.621
latency	1.0	Belief state v5	0.112	0.740	0.365	0.336	-0.572

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods. (continued)

Axis	Level	Method	Success	Track.	Safety	Asym.	Utility
latency	1.0	Oracle state	0.141	0.732	0.365	0.321	-0.540
thermal_drift	0.0	Domain rand.	0.074	0.749	0.359	0.697	-0.677
thermal_drift	0.0	Online ID	0.102	0.725	0.325	0.353	-0.556
thermal_drift	0.0	CBF-FTC MPC	0.070	0.776	0.396	0.365	-0.637
thermal_drift	0.0	UKF observer	0.102	0.723	0.343	0.334	-0.561
thermal_drift	0.0	L1 adaptive	0.079	0.768	0.395	0.318	-0.619
thermal_drift	0.0	Belief state v5	0.123	0.722	0.328	0.295	-0.525
thermal_drift	0.0	Oracle state	0.172	0.704	0.328	0.293	-0.471
thermal_drift	0.2	Domain rand.	0.070	0.755	0.374	0.689	-0.689
thermal_drift	0.2	Online ID	0.083	0.729	0.337	0.350	-0.580
thermal_drift	0.2	CBF-FTC MPC	0.068	0.779	0.413	0.363	-0.648
thermal_drift	0.2	UKF observer	0.097	0.730	0.348	0.334	-0.572
thermal_drift	0.2	L1 adaptive	0.077	0.770	0.395	0.322	-0.621
thermal_drift	0.2	Belief state v5	0.122	0.728	0.343	0.300	-0.537
thermal_drift	0.2	Oracle state	0.148	0.717	0.342	0.297	-0.508
thermal_drift	0.4	Domain rand.	0.068	0.750	0.372	0.701	-0.690
thermal_drift	0.4	Online ID	0.088	0.726	0.336	0.377	-0.580
thermal_drift	0.4	CBF-FTC MPC	0.063	0.778	0.408	0.392	-0.654
thermal_drift	0.4	UKF observer	0.098	0.724	0.337	0.369	-0.569
thermal_drift	0.4	L1 adaptive	0.077	0.758	0.388	0.359	-0.619
thermal_drift	0.4	Belief state v5	0.120	0.723	0.346	0.332	-0.545
thermal_drift	0.4	Oracle state	0.163	0.714	0.338	0.326	-0.495
thermal_drift	0.6	Domain rand.	0.062	0.752	0.360	0.713	-0.692
thermal_drift	0.6	Online ID	0.086	0.730	0.339	0.404	-0.590
thermal_drift	0.6	CBF-FTC MPC	0.071	0.790	0.426	0.419	-0.666
thermal_drift	0.6	UKF observer	0.099	0.735	0.342	0.395	-0.578
thermal_drift	0.6	L1 adaptive	0.086	0.765	0.385	0.384	-0.616
thermal_drift	0.6	Belief state v5	0.108	0.729	0.341	0.363	-0.560
thermal_drift	0.6	Oracle state	0.152	0.718	0.340	0.350	-0.512
thermal_drift	0.8	Domain rand.	0.061	0.754	0.368	0.721	-0.699
thermal_drift	0.8	Online ID	0.084	0.731	0.338	0.438	-0.597
thermal_drift	0.8	CBF-FTC MPC	0.073	0.790	0.425	0.461	-0.670
thermal_drift	0.8	UKF observer	0.110	0.728	0.348	0.429	-0.573
thermal_drift	0.8	L1 adaptive	0.096	0.767	0.394	0.423	-0.618
thermal_drift	0.8	Belief state v5	0.118	0.728	0.351	0.401	-0.562
thermal_drift	0.8	Oracle state	0.158	0.720	0.354	0.384	-0.520
thermal_drift	1.0	Domain rand.	0.066	0.749	0.365	0.782	-0.701
thermal_drift	1.0	Online ID	0.090	0.727	0.340	0.504	-0.603
thermal_drift	1.0	CBF-FTC MPC	0.071	0.786	0.425	0.537	-0.684
thermal_drift	1.0	UKF observer	0.087	0.725	0.334	0.491	-0.598
thermal_drift	1.0	L1 adaptive	0.077	0.766	0.391	0.494	-0.647
thermal_drift	1.0	Belief state v5	0.127	0.729	0.345	0.463	-0.561
thermal_drift	1.0	Oracle state	0.163	0.721	0.349	0.435	-0.522
intermittent_fault	0.0	Domain rand.	0.062	0.749	0.362	0.687	-0.686
intermittent_fault	0.0	Online ID	0.107	0.729	0.338	0.387	-0.565
intermittent_fault	0.0	CBF-FTC MPC	0.073	0.789	0.426	0.399	-0.659
intermittent_fault	0.0	UKF observer	0.095	0.723	0.337	0.379	-0.571
intermittent_fault	0.0	L1 adaptive	0.077	0.755	0.382	0.365	-0.615
intermittent_fault	0.0	Belief state v5	0.128	0.720	0.338	0.349	-0.533
intermittent_fault	0.0	Oracle state	0.184	0.710	0.335	0.338	-0.472
intermittent_fault	0.2	Domain rand.	0.068	0.749	0.357	0.700	-0.680
intermittent_fault	0.2	Online ID	0.092	0.722	0.321	0.391	-0.568
intermittent_fault	0.2	CBF-FTC MPC	0.071	0.781	0.414	0.402	-0.652
intermittent_fault	0.2	UKF observer	0.112	0.723	0.337	0.378	-0.555
intermittent_fault	0.2	L1 adaptive	0.080	0.756	0.376	0.369	-0.611
intermittent_fault	0.2	Belief state v5	0.123	0.726	0.335	0.345	-0.538
intermittent_fault	0.2	Oracle state	0.177	0.709	0.342	0.337	-0.481
intermittent_fault	0.4	Domain rand.	0.067	0.757	0.377	0.704	-0.698
intermittent_fault	0.4	Online ID	0.082	0.731	0.338	0.390	-0.590
intermittent_fault	0.4	CBF-FTC MPC	0.077	0.792	0.436	0.412	-0.666
intermittent_fault	0.4	UKF observer	0.097	0.733	0.343	0.379	-0.578
intermittent_fault	0.4	L1 adaptive	0.079	0.770	0.392	0.371	-0.627
intermittent_fault	0.4	Belief state v5	0.116	0.733	0.348	0.344	-0.556
intermittent_fault	0.4	Oracle state	0.168	0.716	0.343	0.333	-0.495
intermittent_fault	0.6	Domain rand.	0.062	0.744	0.362	0.700	-0.688
intermittent_fault	0.6	Online ID	0.077	0.727	0.341	0.391	-0.596
intermittent_fault	0.6	CBF-FTC MPC	0.067	0.789	0.423	0.408	-0.667
intermittent_fault	0.6	UKF observer	0.086	0.729	0.345	0.385	-0.590
intermittent_fault	0.6	L1 adaptive	0.083	0.761	0.389	0.375	-0.618
intermittent_fault	0.6	Belief state v5	0.126	0.733	0.355	0.351	-0.551
intermittent_fault	0.6	Oracle state	0.158	0.719	0.351	0.344	-0.512
intermittent_fault	0.8	Domain rand.	0.079	0.754	0.361	0.700	-0.675
intermittent_fault	0.8	Online ID	0.082	0.734	0.341	0.396	-0.596
intermittent_fault	0.8	CBF-FTC MPC	0.067	0.785	0.419	0.410	-0.663
intermittent_fault	0.8	UKF observer	0.102	0.725	0.333	0.387	-0.565
intermittent_fault	0.8	L1 adaptive	0.071	0.762	0.382	0.373	-0.626
intermittent_fault	0.8	Belief state v5	0.134	0.724	0.346	0.357	-0.536
intermittent_fault	0.8	Oracle state	0.179	0.708	0.331	0.343	-0.475
intermittent_fault	1.0	Domain rand.	0.063	0.759	0.374	0.704	-0.700
intermittent_fault	1.0	Online ID	0.095	0.733	0.339	0.405	-0.583
intermittent_fault	1.0	CBF-FTC MPC	0.061	0.782	0.418	0.421	-0.669
intermittent_fault	1.0	UKF observer	0.090	0.734	0.348	0.398	-0.591
intermittent_fault	1.0	L1 adaptive	0.079	0.766	0.392	0.384	-0.627
intermittent_fault	1.0	Belief state v5	0.116	0.735	0.354	0.368	-0.565
intermittent_fault	1.0	Oracle state	0.162	0.723	0.355	0.361	-0.516
combined	0.0	Domain rand.	0.148	0.643	0.199	0.076	-0.330
combined	0.0	Online ID	0.159	0.640	0.194	0.031	-0.306
combined	0.0	CBF-FTC MPC	0.137	0.689	0.274	0.031	-0.385
combined	0.0	UKF observer	0.137	0.647	0.209	0.031	-0.338

Table 13: All stress-sweep metrics across six axes, six levels, and seven methods. (continued)

Axis	Level	Method	Success	Track.	Safety	Asym.	Utility
combined	0.0	L1 adaptive	0.127	0.674	0.246	0.029	-0.374
combined	0.0	Belief state v5	0.178	0.656	0.223	0.029	-0.307
combined	0.0	Oracle state	0.212	0.660	0.246	0.034	-0.287
combined	0.2	Domain rand.	0.098	0.680	0.239	0.263	-0.465
combined	0.2	Online ID	0.115	0.660	0.220	0.140	-0.402
combined	0.2	CBF-FTC MPC	0.116	0.710	0.307	0.145	-0.462
combined	0.2	UKF observer	0.130	0.666	0.232	0.141	-0.396
combined	0.2	L1 adaptive	0.111	0.698	0.272	0.136	-0.443
combined	0.2	Belief state v5	0.173	0.667	0.248	0.139	-0.362
combined	0.2	Oracle state	0.191	0.659	0.250	0.152	-0.346
combined	0.4	Domain rand.	0.067	0.727	0.323	0.455	-0.609
combined	0.4	Online ID	0.102	0.705	0.293	0.264	-0.507
combined	0.4	CBF-FTC MPC	0.084	0.763	0.393	0.272	-0.595
combined	0.4	UKF observer	0.121	0.704	0.304	0.264	-0.495
combined	0.4	L1 adaptive	0.094	0.737	0.344	0.254	-0.546
combined	0.4	Belief state v5	0.152	0.703	0.309	0.246	-0.464
combined	0.4	Oracle state	0.190	0.690	0.309	0.245	-0.423
combined	0.6	Domain rand.	0.062	0.759	0.382	0.654	-0.700
combined	0.6	Online ID	0.074	0.738	0.352	0.399	-0.614
combined	0.6	CBF-FTC MPC	0.077	0.793	0.422	0.407	-0.659
combined	0.6	UKF observer	0.102	0.731	0.348	0.391	-0.579
combined	0.6	L1 adaptive	0.086	0.770	0.399	0.381	-0.627
combined	0.6	Belief state v5	0.131	0.730	0.351	0.354	-0.546
combined	0.6	Oracle state	0.158	0.723	0.350	0.345	-0.518
combined	0.8	Domain rand.	0.043	0.820	0.465	0.837	-0.830
combined	0.8	Online ID	0.065	0.793	0.429	0.526	-0.718
combined	0.8	CBF-FTC MPC	0.057	0.845	0.492	0.537	-0.766
combined	0.8	UKF observer	0.076	0.782	0.419	0.513	-0.693
combined	0.8	L1 adaptive	0.062	0.831	0.473	0.505	-0.743
combined	0.8	Belief state v5	0.093	0.782	0.419	0.460	-0.667
combined	0.8	Oracle state	0.124	0.771	0.415	0.451	-0.632
combined	1.0	Domain rand.	0.040	0.854	0.495	1.013	-0.896
combined	1.0	Online ID	0.047	0.854	0.499	0.647	-0.825
combined	1.0	CBF-FTC MPC	0.045	0.885	0.526	0.656	-0.835
combined	1.0	UKF observer	0.054	0.836	0.477	0.623	-0.792
combined	1.0	L1 adaptive	0.053	0.877	0.525	0.615	-0.821
combined	1.0	Belief state v5	0.068	0.833	0.481	0.560	-0.768
combined	1.0	Oracle state	0.099	0.817	0.466	0.552	-0.725

E APPENDIX: COMPLETE FIXED-RISK TABLE

The fixed-risk appendix reports every split, budget, and method. The strict budget 0.05 failure is visible in the full table rather than only in the summary.

Table 14: All fixed-risk deployment rows across both hard splits, four budgets, and six methods.

Split	Budget	Method	Coverage	Succ.	Track.	Safety	Utility
Combined	0.02	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.02	Belief state v5	0.000	0.000	0.000	0.000	0.000
Combined	0.02	Online ID	0.000	0.000	0.000	0.000	0.000
Combined	0.02	Oracle state	0.000	0.000	0.000	0.000	0.000
Combined	0.02	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Combined	0.02	UKF observer	0.000	0.000	0.000	0.000	0.000
Combined	0.05	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Belief state v5	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Online ID	0.000	0.000	0.000	0.000	0.000
Combined	0.05	Oracle state	0.000	0.000	0.000	0.000	0.000
Combined	0.05	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Combined	0.05	UKF observer	0.000	0.000	0.000	0.000	0.000
Combined	0.08	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.08	Belief state v5	0.000	0.000	0.000	0.000	0.000
Combined	0.08	Online ID	0.000	0.000	0.000	0.000	0.000
Combined	0.08	Oracle state	0.000	0.000	0.000	0.000	0.000
Combined	0.08	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Combined	0.08	UKF observer	0.000	0.000	0.000	0.000	0.000
Combined	0.10	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Combined	0.10	Belief state v5	0.000	0.000	0.000	0.000	0.000
Combined	0.10	Online ID	0.000	0.000	0.000	0.000	0.000
Combined	0.10	Oracle state	0.000	0.000	0.000	0.000	0.000
Combined	0.10	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Combined	0.10	UKF observer	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.02	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.02	Belief state v5	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.02	Online ID	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.02	Oracle state	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.02	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.02	UKF observer	0.000	0.000	0.000	0.000	0.000

Table 14: All fixed-risk deployment rows across both hard splits, four budgets, and six methods. (continued)

Split	Budget	Method	Coverage	Succ.	Track.	Safety	Utility
Payload/contact	0.05	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.05	Belief state v5	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.05	Online ID	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.05	Oracle state	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.05	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.05	UKF observer	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.08	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.08	Belief state v5	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.08	Online ID	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.08	Oracle state	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.08	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.08	UKF observer	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.10	CBF-FTC MPC	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.10	Belief state v5	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.10	Online ID	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.10	Oracle state	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.10	L1 adaptive	0.000	0.000	0.000	0.000	0.000
Payload/contact	0.10	UKF observer	0.000	0.000	0.000	0.000	0.000

F APPENDIX: FIXED-RISK PAIRED TESTS

The fixed-risk paired table reports coverage and accepted-episode differences across all budgets. It is included because a deployment reviewer would not accept a single budget-only summary.

Table 15: All fixed-risk paired comparisons for belief state v5 minus each non-v5 reference across budgets.

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
Combined	0.02	CBF-FTC MPC	coverage	0.000	0.000	0.000
Combined	0.02	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Combined	0.02	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Combined	0.02	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Combined	0.02	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Combined	0.02	Online ID	coverage	0.000	0.000	0.000
Combined	0.02	Online ID	accepted_success	0.000	0.000	0.000
Combined	0.02	Online ID	accepted_tracking_error	0.000	0.000	0.000
Combined	0.02	Online ID	accepted_safety	0.000	0.000	0.000
Combined	0.02	Online ID	accepted_utility	0.000	0.000	0.000
Combined	0.02	Oracle state	coverage	0.000	0.000	0.000
Combined	0.02	Oracle state	accepted_success	0.000	0.000	0.000
Combined	0.02	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Combined	0.02	Oracle state	accepted_safety	0.000	0.000	0.000
Combined	0.02	Oracle state	accepted_utility	0.000	0.000	0.000
Combined	0.02	L1 adaptive	coverage	0.000	0.000	0.000
Combined	0.02	L1 adaptive	accepted_success	0.000	0.000	0.000
Combined	0.02	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Combined	0.02	L1 adaptive	accepted_safety	0.000	0.000	0.000
Combined	0.02	L1 adaptive	accepted_utility	0.000	0.000	0.000
Combined	0.02	UKF observer	coverage	0.000	0.000	0.000
Combined	0.02	UKF observer	accepted_success	0.000	0.000	0.000
Combined	0.02	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Combined	0.02	UKF observer	accepted_safety	0.000	0.000	0.000
Combined	0.02	UKF observer	accepted_utility	0.000	0.000	0.000
Combined	0.05	CBF-FTC MPC	coverage	0.000	0.000	0.000
Combined	0.05	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Combined	0.05	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Combined	0.05	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Combined	0.05	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Combined	0.05	Online ID	coverage	0.000	0.000	0.000
Combined	0.05	Online ID	accepted_success	0.000	0.000	0.000
Combined	0.05	Online ID	accepted_tracking_error	0.000	0.000	0.000
Combined	0.05	Online ID	accepted_safety	0.000	0.000	0.000
Combined	0.05	Online ID	accepted_utility	0.000	0.000	0.000
Combined	0.05	Oracle state	coverage	0.000	0.000	0.000
Combined	0.05	Oracle state	accepted_success	0.000	0.000	0.000
Combined	0.05	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Combined	0.05	Oracle state	accepted_safety	0.000	0.000	0.000
Combined	0.05	Oracle state	accepted_utility	0.000	0.000	0.000
Combined	0.05	L1 adaptive	coverage	0.000	0.000	0.000
Combined	0.05	L1 adaptive	accepted_success	0.000	0.000	0.000
Combined	0.05	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Combined	0.05	L1 adaptive	accepted_safety	0.000	0.000	0.000
Combined	0.05	L1 adaptive	accepted_utility	0.000	0.000	0.000
Combined	0.05	UKF observer	coverage	0.000	0.000	0.000
Combined	0.05	UKF observer	accepted_success	0.000	0.000	0.000
Combined	0.05	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Combined	0.05	UKF observer	accepted_safety	0.000	0.000	0.000
Combined	0.05	UKF observer	accepted_utility	0.000	0.000	0.000

Table 15: All fixed-risk paired comparisons for belief state v5 minus each non-v5 reference across budgets. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
Combined	0.08	CBF-FTC MPC	coverage	0.000	0.000	0.000
Combined	0.08	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Combined	0.08	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Combined	0.08	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Combined	0.08	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Combined	0.08	Online ID	coverage	0.000	0.000	0.000
Combined	0.08	Online ID	accepted_success	0.000	0.000	0.000
Combined	0.08	Online ID	accepted_tracking_error	0.000	0.000	0.000
Combined	0.08	Online ID	accepted_safety	0.000	0.000	0.000
Combined	0.08	Online ID	accepted_utility	0.000	0.000	0.000
Combined	0.08	Oracle state	coverage	0.000	0.000	0.000
Combined	0.08	Oracle state	accepted_success	0.000	0.000	0.000
Combined	0.08	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Combined	0.08	Oracle state	accepted_safety	0.000	0.000	0.000
Combined	0.08	Oracle state	accepted_utility	0.000	0.000	0.000
Combined	0.08	L1 adaptive	coverage	0.000	0.000	0.000
Combined	0.08	L1 adaptive	accepted_success	0.000	0.000	0.000
Combined	0.08	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Combined	0.08	L1 adaptive	accepted_safety	0.000	0.000	0.000
Combined	0.08	L1 adaptive	accepted_utility	0.000	0.000	0.000
Combined	0.08	UKF observer	coverage	0.000	0.000	0.000
Combined	0.08	UKF observer	accepted_success	0.000	0.000	0.000
Combined	0.08	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Combined	0.08	UKF observer	accepted_safety	0.000	0.000	0.000
Combined	0.08	UKF observer	accepted_utility	0.000	0.000	0.000
Combined	0.10	CBF-FTC MPC	coverage	0.000	0.000	0.000
Combined	0.10	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Combined	0.10	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Combined	0.10	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Combined	0.10	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Combined	0.10	Online ID	coverage	0.000	0.000	0.000
Combined	0.10	Online ID	accepted_success	0.000	0.000	0.000
Combined	0.10	Online ID	accepted_tracking_error	0.000	0.000	0.000
Combined	0.10	Online ID	accepted_safety	0.000	0.000	0.000
Combined	0.10	Online ID	accepted_utility	0.000	0.000	0.000
Combined	0.10	Oracle state	coverage	0.000	0.000	0.000
Combined	0.10	Oracle state	accepted_success	0.000	0.000	0.000
Combined	0.10	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Combined	0.10	Oracle state	accepted_safety	0.000	0.000	0.000
Combined	0.10	Oracle state	accepted_utility	0.000	0.000	0.000
Combined	0.10	L1 adaptive	coverage	0.000	0.000	0.000
Combined	0.10	L1 adaptive	accepted_success	0.000	0.000	0.000
Combined	0.10	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Combined	0.10	L1 adaptive	accepted_safety	0.000	0.000	0.000
Combined	0.10	L1 adaptive	accepted_utility	0.000	0.000	0.000
Combined	0.10	UKF observer	coverage	0.000	0.000	0.000
Combined	0.10	UKF observer	accepted_success	0.000	0.000	0.000
Combined	0.10	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Combined	0.10	UKF observer	accepted_safety	0.000	0.000	0.000
Combined	0.10	UKF observer	accepted_utility	0.000	0.000	0.000
Payload/contact	0.02	CBF-FTC MPC	coverage	0.000	0.000	0.000
Payload/contact	0.02	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Payload/contact	0.02	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.02	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Payload/contact	0.02	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Payload/contact	0.02	Online ID	coverage	0.000	0.000	0.000
Payload/contact	0.02	Online ID	accepted_success	0.000	0.000	0.000
Payload/contact	0.02	Online ID	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.02	Online ID	accepted_safety	0.000	0.000	0.000
Payload/contact	0.02	Online ID	accepted_utility	0.000	0.000	0.000
Payload/contact	0.02	Oracle state	coverage	0.000	0.000	0.000
Payload/contact	0.02	Oracle state	accepted_success	0.000	0.000	0.000
Payload/contact	0.02	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.02	Oracle state	accepted_safety	0.000	0.000	0.000
Payload/contact	0.02	Oracle state	accepted_utility	0.000	0.000	0.000
Payload/contact	0.02	L1 adaptive	coverage	0.000	0.000	0.000
Payload/contact	0.02	L1 adaptive	accepted_success	0.000	0.000	0.000
Payload/contact	0.02	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.02	L1 adaptive	accepted_safety	0.000	0.000	0.000
Payload/contact	0.02	L1 adaptive	accepted_utility	0.000	0.000	0.000
Payload/contact	0.02	UKF observer	coverage	0.000	0.000	0.000
Payload/contact	0.02	UKF observer	accepted_success	0.000	0.000	0.000
Payload/contact	0.02	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.02	UKF observer	accepted_safety	0.000	0.000	0.000
Payload/contact	0.02	UKF observer	accepted_utility	0.000	0.000	0.000
Payload/contact	0.05	CBF-FTC MPC	coverage	0.000	0.000	0.000
Payload/contact	0.05	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Payload/contact	0.05	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.05	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Payload/contact	0.05	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Payload/contact	0.05	Online ID	coverage	0.000	0.000	0.000
Payload/contact	0.05	Online ID	accepted_success	0.000	0.000	0.000
Payload/contact	0.05	Online ID	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.05	Online ID	accepted_safety	0.000	0.000	0.000
Payload/contact	0.05	Online ID	accepted_utility	0.000	0.000	0.000
Payload/contact	0.05	Oracle state	coverage	0.000	0.000	0.000
Payload/contact	0.05	Oracle state	accepted_success	0.000	0.000	0.000
Payload/contact	0.05	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.05	Oracle state	accepted_safety	0.000	0.000	0.000

Table 15: All fixed-risk paired comparisons for belief state v5 minus each non-v5 reference across budgets. (continued)

Split	Budget	Reference	Metric	Mean diff	CI95	Lower95
Payload/contact	0.05	Oracle state	accepted_utility	0.000	0.000	0.000
Payload/contact	0.05	L1 adaptive	coverage	0.000	0.000	0.000
Payload/contact	0.05	L1 adaptive	accepted_success	0.000	0.000	0.000
Payload/contact	0.05	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.05	L1 adaptive	accepted_safety	0.000	0.000	0.000
Payload/contact	0.05	L1 adaptive	accepted_utility	0.000	0.000	0.000
Payload/contact	0.05	UKF observer	coverage	0.000	0.000	0.000
Payload/contact	0.05	UKF observer	accepted_success	0.000	0.000	0.000
Payload/contact	0.05	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.05	UKF observer	accepted_safety	0.000	0.000	0.000
Payload/contact	0.05	UKF observer	accepted_utility	0.000	0.000	0.000
Payload/contact	0.08	CBF-FTC MPC	coverage	0.000	0.000	0.000
Payload/contact	0.08	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Payload/contact	0.08	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.08	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Payload/contact	0.08	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Payload/contact	0.08	Online ID	coverage	0.000	0.000	0.000
Payload/contact	0.08	Online ID	accepted_success	0.000	0.000	0.000
Payload/contact	0.08	Online ID	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.08	Online ID	accepted_safety	0.000	0.000	0.000
Payload/contact	0.08	Online ID	accepted_utility	0.000	0.000	0.000
Payload/contact	0.08	Oracle state	coverage	0.000	0.000	0.000
Payload/contact	0.08	Oracle state	accepted_success	0.000	0.000	0.000
Payload/contact	0.08	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.08	Oracle state	accepted_safety	0.000	0.000	0.000
Payload/contact	0.08	Oracle state	accepted_utility	0.000	0.000	0.000
Payload/contact	0.08	L1 adaptive	coverage	0.000	0.000	0.000
Payload/contact	0.08	L1 adaptive	accepted_success	0.000	0.000	0.000
Payload/contact	0.08	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.08	L1 adaptive	accepted_safety	0.000	0.000	0.000
Payload/contact	0.08	L1 adaptive	accepted_utility	0.000	0.000	0.000
Payload/contact	0.08	UKF observer	coverage	0.000	0.000	0.000
Payload/contact	0.08	UKF observer	accepted_success	0.000	0.000	0.000
Payload/contact	0.08	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.08	UKF observer	accepted_safety	0.000	0.000	0.000
Payload/contact	0.08	UKF observer	accepted_utility	0.000	0.000	0.000
Payload/contact	0.10	CBF-FTC MPC	coverage	0.000	0.000	0.000
Payload/contact	0.10	CBF-FTC MPC	accepted_success	0.000	0.000	0.000
Payload/contact	0.10	CBF-FTC MPC	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.10	CBF-FTC MPC	accepted_safety	0.000	0.000	0.000
Payload/contact	0.10	CBF-FTC MPC	accepted_utility	0.000	0.000	0.000
Payload/contact	0.10	Online ID	coverage	0.000	0.000	0.000
Payload/contact	0.10	Online ID	accepted_success	0.000	0.000	0.000
Payload/contact	0.10	Online ID	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.10	Online ID	accepted_safety	0.000	0.000	0.000
Payload/contact	0.10	Online ID	accepted_utility	0.000	0.000	0.000
Payload/contact	0.10	Oracle state	coverage	0.000	0.000	0.000
Payload/contact	0.10	Oracle state	accepted_success	0.000	0.000	0.000
Payload/contact	0.10	Oracle state	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.10	Oracle state	accepted_safety	0.000	0.000	0.000
Payload/contact	0.10	Oracle state	accepted_utility	0.000	0.000	0.000
Payload/contact	0.10	L1 adaptive	coverage	0.000	0.000	0.000
Payload/contact	0.10	L1 adaptive	accepted_success	0.000	0.000	0.000
Payload/contact	0.10	L1 adaptive	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.10	L1 adaptive	accepted_safety	0.000	0.000	0.000
Payload/contact	0.10	L1 adaptive	accepted_utility	0.000	0.000	0.000
Payload/contact	0.10	UKF observer	coverage	0.000	0.000	0.000
Payload/contact	0.10	UKF observer	accepted_success	0.000	0.000	0.000
Payload/contact	0.10	UKF observer	accepted_tracking_error	0.000	0.000	0.000
Payload/contact	0.10	UKF observer	accepted_safety	0.000	0.000	0.000
Payload/contact	0.10	UKF observer	accepted_utility	0.000	0.000	0.000

G APPENDIX: ABLATION SEED UNITS

The mechanism gate was evaluated on seed means from payload/contact and combined hard shift. The full mechanism does not achieve the required clean robust-utility margin over all ablations.

Table 16: Ablation seed means for the two hard ablation splits.

Split	Ablation	Seed	Success	Tracking	Safety	Utility
Combined	Full belief v5	0	0.083	0.735	0.341	-0.590
Combined	Full belief v5	1	0.131	0.730	0.344	-0.534
Combined	Full belief v5	2	0.131	0.707	0.319	-0.508
Combined	Full belief v5	3	0.119	0.730	0.346	-0.551
Combined	Full belief v5	4	0.107	0.731	0.365	-0.574
Combined	Full belief v5	5	0.083	0.715	0.332	-0.575
Combined	Full belief v5	6	0.143	0.698	0.286	-0.479

Table 16: Ablation seed means for the two hard ablation splits. (continued)

Split	Ablation	Seed	Success	Tracking	Safety	Utility
Combined	Full belief v5	7	0.125	0.751	0.387	-0.573
Combined	Full belief v5	8	0.125	0.753	0.379	-0.576
Combined	Full belief v5	9	0.125	0.741	0.372	-0.561
Combined	- deadzone/sat. state	0	0.048	0.766	0.374	-0.652
Combined	- deadzone/sat. state	1	0.077	0.755	0.360	-0.603
Combined	- deadzone/sat. state	2	0.095	0.729	0.353	-0.568
Combined	- deadzone/sat. state	3	0.077	0.751	0.372	-0.611
Combined	- deadzone/sat. state	4	0.071	0.772	0.399	-0.642
Combined	- deadzone/sat. state	5	0.077	0.743	0.347	-0.596
Combined	- deadzone/sat. state	6	0.113	0.726	0.326	-0.533
Combined	- deadzone/sat. state	7	0.065	0.760	0.385	-0.628
Combined	- deadzone/sat. state	8	0.089	0.773	0.396	-0.621
Combined	- deadzone/sat. state	9	0.107	0.745	0.390	-0.583
Combined	- latency comp.	0	0.089	0.730	0.336	-0.581
Combined	- latency comp.	1	0.119	0.733	0.351	-0.551
Combined	- latency comp.	2	0.095	0.716	0.337	-0.559
Combined	- latency comp.	3	0.065	0.747	0.365	-0.624
Combined	- latency comp.	4	0.107	0.748	0.386	-0.592
Combined	- latency comp.	5	0.071	0.736	0.335	-0.599
Combined	- latency comp.	6	0.101	0.741	0.347	-0.571
Combined	- latency comp.	7	0.107	0.734	0.349	-0.560
Combined	- latency comp.	8	0.095	0.753	0.370	-0.602
Combined	- latency comp.	9	0.131	0.741	0.378	-0.559
Combined	- residual update	0	0.077	0.748	0.372	-0.624
Combined	- residual update	1	0.125	0.727	0.332	-0.542
Combined	- residual update	2	0.071	0.723	0.345	-0.599
Combined	- residual update	3	0.065	0.738	0.364	-0.623
Combined	- residual update	4	0.077	0.753	0.386	-0.633
Combined	- residual update	5	0.077	0.739	0.355	-0.611
Combined	- residual update	6	0.071	0.740	0.349	-0.610
Combined	- residual update	7	0.107	0.751	0.386	-0.599
Combined	- residual update	8	0.054	0.729	0.334	-0.617
Combined	- residual update	9	0.089	0.740	0.373	-0.606
Combined	- persistent memory	0	0.071	0.764	0.415	-0.659
Combined	- persistent memory	1	0.083	0.739	0.357	-0.601
Combined	- persistent memory	2	0.101	0.720	0.345	-0.567
Combined	- persistent memory	3	0.101	0.748	0.364	-0.591
Combined	- persistent memory	4	0.065	0.763	0.404	-0.655
Combined	- persistent memory	5	0.083	0.731	0.346	-0.594
Combined	- persistent memory	6	0.089	0.734	0.355	-0.591
Combined	- persistent memory	7	0.071	0.759	0.380	-0.633
Combined	- persistent memory	8	0.119	0.743	0.355	-0.567
Combined	- persistent memory	9	0.071	0.748	0.371	-0.623
Combined	- safety clipping	0	0.089	0.682	0.448	-0.641
Combined	- safety clipping	1	0.119	0.664	0.392	-0.566
Combined	- safety clipping	2	0.125	0.667	0.417	-0.575
Combined	- safety clipping	3	0.060	0.690	0.423	-0.657
Combined	- safety clipping	4	0.095	0.691	0.468	-0.646
Combined	- safety clipping	5	0.125	0.682	0.419	-0.589
Combined	- safety clipping	6	0.095	0.694	0.450	-0.637
Combined	- safety clipping	7	0.131	0.672	0.409	-0.563
Combined	- safety clipping	8	0.113	0.671	0.427	-0.601
Combined	- safety clipping	9	0.101	0.694	0.477	-0.647
Combined	- sign-specific asym.	0	0.077	0.768	0.390	-0.634
Combined	- sign-specific asym.	1	0.095	0.744	0.352	-0.577
Combined	- sign-specific asym.	2	0.065	0.742	0.358	-0.607
Combined	- sign-specific asym.	3	0.095	0.746	0.387	-0.599
Combined	- sign-specific asym.	4	0.083	0.790	0.448	-0.666
Combined	- sign-specific asym.	5	0.089	0.743	0.361	-0.590
Combined	- sign-specific asym.	6	0.077	0.735	0.342	-0.586
Combined	- sign-specific asym.	7	0.071	0.757	0.369	-0.614
Combined	- sign-specific asym.	8	0.107	0.759	0.369	-0.585
Combined	- sign-specific asym.	9	0.095	0.737	0.383	-0.591
Combined	- drift update	0	0.113	0.738	0.371	-0.577
Combined	- drift update	1	0.137	0.711	0.334	-0.512
Combined	- drift update	2	0.101	0.707	0.315	-0.539
Combined	- drift update	3	0.155	0.713	0.335	-0.499
Combined	- drift update	4	0.101	0.751	0.385	-0.598
Combined	- drift update	5	0.107	0.709	0.326	-0.544
Combined	- drift update	6	0.077	0.729	0.354	-0.594
Combined	- drift update	7	0.083	0.749	0.381	-0.611
Combined	- drift update	8	0.101	0.736	0.365	-0.584
Combined	- drift update	9	0.107	0.732	0.358	-0.568
Combined	No belief uncertainty	0	0.083	0.730	0.365	-0.602
Combined	No belief uncertainty	1	0.101	0.725	0.332	-0.556
Combined	No belief uncertainty	2	0.137	0.695	0.335	-0.507
Combined	No belief uncertainty	3	0.125	0.722	0.344	-0.541
Combined	No belief uncertainty	4	0.143	0.757	0.411	-0.573
Combined	No belief uncertainty	5	0.089	0.739	0.350	-0.590
Combined	No belief uncertainty	6	0.083	0.721	0.324	-0.570
Combined	No belief uncertainty	7	0.089	0.732	0.349	-0.580
Combined	No belief uncertainty	8	0.131	0.743	0.364	-0.558
Combined	No belief uncertainty	9	0.113	0.729	0.383	-0.577
Combined	Scalar-only latent gain	0	0.083	0.756	0.391	-0.622
Combined	Scalar-only latent gain	1	0.065	0.752	0.365	-0.617
Combined	Scalar-only latent gain	2	0.107	0.741	0.373	-0.576
Combined	Scalar-only latent gain	3	0.095	0.755	0.362	-0.587
Combined	Scalar-only latent gain	4	0.089	0.763	0.397	-0.622
Combined	Scalar-only latent gain	5	0.065	0.760	0.375	-0.629

Table 16: Ablation seed means for the two hard ablation splits. (continued)

Split	Ablation	Seed	Success	Tracking	Safety	Utility
Combined	Scalar-only latent gain	6	0.095	0.754	0.362	-0.588
Combined	Scalar-only latent gain	7	0.065	0.753	0.372	-0.621
Combined	Scalar-only latent gain	8	0.089	0.755	0.365	-0.598
Combined	Scalar-only latent gain	9	0.048	0.766	0.404	-0.662
Payload/contact	Full belief v5	0	0.077	0.735	0.370	-0.574
Payload/contact	Full belief v5	1	0.185	0.682	0.285	-0.396
Payload/contact	Full belief v5	2	0.190	0.695	0.312	-0.409
Payload/contact	Full belief v5	3	0.149	0.698	0.310	-0.451
Payload/contact	Full belief v5	4	0.107	0.704	0.294	-0.491
Payload/contact	Full belief v5	5	0.143	0.683	0.279	-0.432
Payload/contact	Full belief v5	6	0.149	0.703	0.291	-0.447
Payload/contact	Full belief v5	7	0.196	0.679	0.280	-0.379
Payload/contact	Full belief v5	8	0.161	0.671	0.263	-0.401
Payload/contact	Full belief v5	9	0.137	0.658	0.232	-0.404
Payload/contact	- deadzone/sat. state	0	0.083	0.746	0.359	-0.564
Payload/contact	- deadzone/sat. state	1	0.119	0.689	0.276	-0.448
Payload/contact	- deadzone/sat. state	2	0.095	0.733	0.344	-0.529
Payload/contact	- deadzone/sat. state	3	0.113	0.692	0.294	-0.468
Payload/contact	- deadzone/sat. state	4	0.119	0.733	0.348	-0.511
Payload/contact	- deadzone/sat. state	5	0.131	0.712	0.314	-0.473
Payload/contact	- deadzone/sat. state	6	0.083	0.723	0.332	-0.539
Payload/contact	- deadzone/sat. state	7	0.131	0.706	0.304	-0.464
Payload/contact	- deadzone/sat. state	8	0.125	0.697	0.278	-0.448
Payload/contact	- deadzone/sat. state	9	0.149	0.688	0.261	-0.415
Payload/contact	- latency comp.	0	0.113	0.732	0.355	-0.529
Payload/contact	- latency comp.	1	0.125	0.689	0.298	-0.463
Payload/contact	- latency comp.	2	0.101	0.702	0.297	-0.491
Payload/contact	- latency comp.	3	0.095	0.688	0.280	-0.482
Payload/contact	- latency comp.	4	0.131	0.720	0.337	-0.494
Payload/contact	- latency comp.	5	0.155	0.691	0.290	-0.431
Payload/contact	- latency comp.	6	0.113	0.708	0.310	-0.494
Payload/contact	- latency comp.	7	0.143	0.667	0.264	-0.417
Payload/contact	- latency comp.	8	0.161	0.666	0.271	-0.401
Payload/contact	- latency comp.	9	0.113	0.670	0.252	-0.441
Payload/contact	- residual update	0	0.065	0.726	0.357	-0.584
Payload/contact	- residual update	1	0.131	0.685	0.288	-0.457
Payload/contact	- residual update	2	0.131	0.712	0.308	-0.480
Payload/contact	- residual update	3	0.167	0.683	0.297	-0.425
Payload/contact	- residual update	4	0.101	0.719	0.314	-0.517
Payload/contact	- residual update	5	0.048	0.698	0.304	-0.557
Payload/contact	- residual update	6	0.077	0.713	0.287	-0.528
Payload/contact	- residual update	7	0.101	0.680	0.286	-0.484
Payload/contact	- residual update	8	0.083	0.684	0.281	-0.501
Payload/contact	- residual update	9	0.149	0.675	0.274	-0.430
Payload/contact	- persistent memory	0	0.107	0.727	0.336	-0.531
Payload/contact	- persistent memory	1	0.113	0.711	0.321	-0.506
Payload/contact	- persistent memory	2	0.131	0.707	0.320	-0.483
Payload/contact	- persistent memory	3	0.095	0.708	0.304	-0.509
Payload/contact	- persistent memory	4	0.101	0.697	0.286	-0.492
Payload/contact	- persistent memory	5	0.137	0.692	0.300	-0.461
Payload/contact	- persistent memory	6	0.119	0.698	0.287	-0.476
Payload/contact	- persistent memory	7	0.137	0.657	0.262	-0.423
Payload/contact	- persistent memory	8	0.089	0.673	0.258	-0.476
Payload/contact	- persistent memory	9	0.161	0.668	0.252	-0.402
Payload/contact	- safety clipping	0	0.155	0.659	0.295	-0.434
Payload/contact	- safety clipping	1	0.137	0.647	0.262	-0.425
Payload/contact	- safety clipping	2	0.208	0.654	0.254	-0.350
Payload/contact	- safety clipping	3	0.173	0.648	0.250	-0.382
Payload/contact	- safety clipping	4	0.089	0.670	0.293	-0.502
Payload/contact	- safety clipping	5	0.190	0.666	0.287	-0.393
Payload/contact	- safety clipping	6	0.155	0.655	0.261	-0.415
Payload/contact	- safety clipping	7	0.173	0.631	0.221	-0.358
Payload/contact	- safety clipping	8	0.149	0.637	0.222	-0.384
Payload/contact	- safety clipping	9	0.167	0.635	0.218	-0.366
Payload/contact	- sign-specific asym.	0	0.089	0.734	0.351	-0.547
Payload/contact	- sign-specific asym.	1	0.125	0.681	0.269	-0.436
Payload/contact	- sign-specific asym.	2	0.054	0.725	0.340	-0.566
Payload/contact	- sign-specific asym.	3	0.119	0.701	0.294	-0.467
Payload/contact	- sign-specific asym.	4	0.083	0.718	0.320	-0.525
Payload/contact	- sign-specific asym.	5	0.125	0.700	0.316	-0.475
Payload/contact	- sign-specific asym.	6	0.060	0.725	0.330	-0.563
Payload/contact	- sign-specific asym.	7	0.161	0.697	0.302	-0.430
Payload/contact	- sign-specific asym.	8	0.095	0.691	0.280	-0.479
Payload/contact	- sign-specific asym.	9	0.119	0.680	0.251	-0.436
Payload/contact	- drift update	0	0.125	0.724	0.336	-0.502
Payload/contact	- drift update	1	0.113	0.686	0.284	-0.468
Payload/contact	- drift update	2	0.119	0.689	0.303	-0.470
Payload/contact	- drift update	3	0.107	0.680	0.287	-0.469
Payload/contact	- drift update	4	0.143	0.706	0.320	-0.465
Payload/contact	- drift update	5	0.107	0.710	0.327	-0.506
Payload/contact	- drift update	6	0.089	0.706	0.299	-0.509
Payload/contact	- drift update	7	0.149	0.679	0.267	-0.420
Payload/contact	- drift update	8	0.113	0.682	0.272	-0.461
Payload/contact	- drift update	9	0.137	0.659	0.239	-0.408
Payload/contact	No belief uncertainty	0	0.077	0.725	0.359	-0.567
Payload/contact	No belief uncertainty	1	0.107	0.687	0.287	-0.477
Payload/contact	No belief uncertainty	2	0.173	0.691	0.301	-0.418
Payload/contact	No belief uncertainty	3	0.131	0.682	0.291	-0.451
Payload/contact	No belief uncertainty	4	0.107	0.701	0.312	-0.497

Table 16: Ablation seed means for the two hard ablation splits. (continued)

Split	Ablation	Seed	Success	Tracking	Safety	Utility
Payload/contact	No belief uncertainty	5	0.155	0.698	0.299	-0.438
Payload/contact	No belief uncertainty	6	0.119	0.702	0.309	-0.486
Payload/contact	No belief uncertainty	7	0.173	0.683	0.265	-0.399
Payload/contact	No belief uncertainty	8	0.155	0.676	0.282	-0.421
Payload/contact	No belief uncertainty	9	0.143	0.666	0.246	-0.409
Payload/contact	Scalar-only latent gain	0	0.083	0.744	0.370	-0.570
Payload/contact	Scalar-only latent gain	1	0.089	0.705	0.301	-0.502
Payload/contact	Scalar-only latent gain	2	0.101	0.728	0.340	-0.519
Payload/contact	Scalar-only latent gain	3	0.089	0.704	0.284	-0.491
Payload/contact	Scalar-only latent gain	4	0.101	0.761	0.362	-0.547
Payload/contact	Scalar-only latent gain	5	0.083	0.716	0.306	-0.518
Payload/contact	Scalar-only latent gain	6	0.083	0.722	0.305	-0.524
Payload/contact	Scalar-only latent gain	7	0.125	0.700	0.300	-0.465
Payload/contact	Scalar-only latent gain	8	0.113	0.686	0.293	-0.465
Payload/contact	Scalar-only latent gain	9	0.125	0.680	0.276	-0.442

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